

Local Immigration Enforcement and Medicare: Health-System Spillovers from County Sanctuary Policies

Thesis Chapter Draft

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Abstract

This chapter examines whether county restrictions on cooperation with federal immigration detainers spill over into Medicare spending and healthcare utilization. It links exact sanctuary-policy dates for 296 large U.S. counties to Medicare records from 2007 through 2024, three independent sources on the care workforce, and a separate administrative first-stage sample constructed from 1.8 million ICE detainer records. Sanctuary adoption materially disrupts the local jail-to-ICE pipeline: recorded local declines increase and the probability that a detainer produces federal custody falls sharply, while ICE notification requests remain approximately unchanged. Medicare responds more gradually. Standardized spending per beneficiary rises by 2.2 percent after five years and 4.5 percent after eight years. At year eight, inpatient participation rises by 0.53 percentage points, covered inpatient days rise by 125.8 per 1,000 beneficiaries, and average stays lengthen by 0.263 days. These inpatient outcomes have clean pretreatment profiles and survive within-family false-discovery-rate adjustment, whereas emergency visits, ambulance use, and readmissions do not increase detectably. The chapter then evaluates two candidate pathways. QCEW workforce estimates do not replicate in County Business Patterns or public ACS occupation-by-citizenship data. Public Medicare outpatient measures also fail to show a coherent access mechanism: outpatient participation and evaluation-and-management events fall modestly, testing increases, and the strongest FQHC and imaging patterns begin before adoption. Comparability-adjusted support is strongest for 2014 adopters; the 2013 cohort lacks common support and remains sensitive to California. The evidence establishes a delayed inpatient-utilization and federal-spending spillover following a verified change in local enforcement cooperation, but it does not determine whether the additional care reflects access, accumulated health needs, provider behavior, coding, or beneficiary composition.

JEL classifications: I11, I18, J15, J61, J68

Keywords: sanctuary policies; immigration enforcement; Medicare; inpatient utilization; policy spillovers; difference in differences

1 Introduction

Immigration policy can affect health markets even when it does not change health-insurance eligibility or provider payment. Local enforcement institutions can alter household stability, trust in public institutions, informal caregiving, and the supply of labor used by health and household-service organizations. These changes may affect when older residents enter care and how intensively they use covered services. Such spillovers matter for the incidence of immigration enforcement because Medicare beneficiaries are largely outside the population directly targeted by that enforcement. Evidence linking local enforcement institutions to Medicare utilization nevertheless remains limited.

We study county sanctuary policies: formal limits on local compliance with federal immigration detainer requests. Hausman (2020) shows that these policies reduced deportations by roughly one third without increasing crime in a sample of counties with large Hispanic populations. The policy changes the local reach and salience of immigration enforcement but does not expand Medicare eligibility. A response among Medicare beneficiaries must therefore operate through household, community, workforce, provider, or compositional spillovers. Greater institutional contact could increase measured use by revealing unmet need; delayed presentation could increase admission intensity; and a change in care labor could either complement hospital care or substitute toward home-based services. The sign and location of the spending response are consequently empirical questions.

Our empirical analysis joins county policy dates to annual employment, establishment, payroll, wage, population, socioeconomic, and Medicare outcomes. The balanced design frame contains 296 counties from 2007 through 2024; 148 adopt a policy and 148 never adopt during the sample. Adoption is highly concentrated: 126 of 148 treated counties adopt in 2013 or 2014. We estimate dynamic treatment effects with the imputation estimator of Borusyak, Jaravel, and Spiess (2024), which avoids the negative weighting and contamination problems of conventional two-way fixed-effects event studies under staggered adoption. County and year fixed effects define untreated potential outcomes, and inference is clustered by state.

The principal result is a delayed increase in standardized Medicare spending per beneficiary. The point estimate is essentially zero in the adoption year, grows to 1.3 log points in year three, and reaches 2.2 log points in year five and 4.4 log points in year eight. Six pretreatment coefficients are jointly insignificant ($p = 0.183$). Three independently reported inpatient margins locate the change. At year eight, inpatient-user share rises by 0.53 percentage points, covered days rise by 125.8 per 1,000 beneficiaries, and covered days per stay rise by 0.263. Their pretreatment p values range from 0.481 to 0.923, and all remain significant after adjustment within the acute-access outcome family. The long delay is consistent with gradual changes

in health, institutional contact, referral patterns, or utilization, though the aggregate data cannot distinguish those explanations.

The spending result survives several consequential changes. Assigning policies adopted from July onward to the following calendar year yields a year-five effect of 2.3 percent and a year-eight effect of 4.9 percent. Moving only October-through-December policies yields 2.4 and 5.0 percent. Adding log population gives 2.4 percent at year five, and omitting the pandemic years gives 2.5 percent. Excluding California reduces the estimates to 1.7 percent in year five and 2.7 percent in year eight, both significant only at the ten or five percent boundary. The California sensitivity is economically meaningful because adoption was concentrated in that state and should temper claims of national representativeness.

Evidence on a care-workforce mechanism is mixed and does not support mediation. QCEW employment and establishment measures increase, but matched County Business Patterns measures do not replicate them. We therefore add an independent public ACS EEO test comparing detailed occupation counts in 2006–2010 and 2014–2018 for 2013/2014 adopters and never-treated counties. The adjusted number of core aides rises by 0.088 log points, but the effect is concentrated in the 2014 cohort, does not survive adjustment across ten ACS outcomes, and is not accompanied by a significant increase per older resident. Neither noncitizen aide employment nor the noncitizen share changes significantly. These results reject a simple account in which sanctuary adoption produces a readily detectable expansion of the noncitizen direct-care workforce.

This chapter makes four contributions. First, it documents a spillover from local immigration-enforcement institutions to a federal insurance program serving a population largely outside the directly targeted immigrant population. This perspective complements Almuhausen, Amuedo-Dorantes, and Furtado (2024), who show that Secure Communities increased institutionalization among older Americans. Second, it verifies within the study setting that the legal treatment changed administrative behavior. A separate reconstruction of the ICE detainer pipeline shows that adoption increased recorded local declines and sharply reduced the conversion of detainees into federal custody. Third, it decomposes the delayed spending pattern into a wider inpatient-user margin and greater intensity per stay and distinguishes that pattern from emergency use, ambulance events, and readmissions. Fourth, it subjects two intuitive explanations—care-workforce capacity and outpatient access—to unusually broad measurement checks. The disagreement across QCEW, County Business Patterns, and ACS workforce data, together with mixed outpatient-use patterns, narrows the interpretation rather than supplying a convenient mechanism. Cohort-specific risk sets further show that comparability-adjusted support is strongest for 2014 and inadequate for 2013.

The paper is deliberately narrower than a welfare assessment. Standardized Medicare spending per beneficiary is the sole primary outcome. Inpatient participation, covered days, and length of stay form the

primary mechanism family used to locate the spending response. Other utilization measures form a secondary falsification family, while employment, establishments, wages, provider capacity, and ACS occupation measures are exploratory mechanisms. The estimates omit privately paid household labor, informal care, Medicaid long-term services and supports, and the welfare of immigrant households. Differential trends in Medicare Advantage participation and dual eligibility also make beneficiary composition a material threat. Our preferred interpretation is a reduced-form spending and inpatient-utilization response with stronger comparability-adjusted support for 2014 than for 2013.

2 Institutional setting and conceptual framework

2.1 Immigration detainers and county sanctuary policies

Federal immigration detainers ask a state or local law-enforcement agency to hold a person beyond the time at which the agency would otherwise release that person, giving federal officers an opportunity to assume custody. Local participation makes routine contact with the criminal-justice system an input into federal removal. A sanctuary policy in this study is a county rule limiting compliance with those requests. The treatment is thus a change in local cooperation, not a guarantee against federal enforcement and not a uniform package of services.

The policy dates come from Hausman's county policy file and refer to the first qualifying restriction. Hausman documents a large decline in deportations following adoption and no corresponding increase in reported crime. Those findings motivate the first-stage interpretation used here: adoption lowers the expected reach of immigration enforcement in the county. Our annual treatment indicator switches on in the calendar year of the recorded policy date and remains on. Because many policies begin late in the year, we also assign July-through-December and October-through-December adoptions to the next year.

2.2 The evolution of interior enforcement

Interior immigration enforcement in the United States is institutionally decentralized. Federal agencies possess removal authority, but the practical reach of enforcement often depends on information and custody supplied by state and local criminal-justice agencies. Programs such as Secure Communities automated the comparison of fingerprints taken after a local arrest with federal immigration databases. Detainers then allowed federal officers to request continued custody. This structure made local cooperation consequential even though counties could not change federal immigration law.

The legal and political environment changed rapidly during the study period. Litigation questioned whether prolonged detention without a judicial warrant exposed local governments to constitutional liability. At the same time, immigrant advocates argued that detainer cooperation weakened trust in police and extended immigration enforcement to people arrested for minor offenses or never convicted. Counties responded through ordinances, sheriff directives, and administrative policies that limited the circumstances in which a detainer would be honored. These rules varied in detail, but each reduced local participation relative to the prior regime.

A county policy may operate through more than completed removals. The expected probability of questioning, detention, or family separation can affect behavior before any enforcement encounter occurs. Workers may avoid driving, change residences, reduce labor-force participation, or move into informal employment. Employers may face higher turnover and recruiting costs. Protective policies can reverse some of these behavioral responses by changing perceived as well as realized risk. This distinction matters because a relatively small number of directly affected workers can generate broader equilibrium effects in a thin local care market.

Sanctuary is also an imprecise political label. The policies coded here specifically concern detainer cooperation. They do not necessarily restrict information sharing in every setting, provide eligibility for public benefits, or prevent federal operations. The empirical treatment should therefore be interpreted as a local institutional constraint on one enforcement channel. The resulting estimates average over heterogeneous legal provisions and implementation intensity.

2.3 Care labor, utilization, and spending

A reduction in enforcement risk can affect care markets through labor supply. Workers may be more willing to remain in the county, accept formal employment, increase hours, or move from informal arrangements into payroll employment. Employers may enter or expand if recruiting becomes easier. These responses predict higher employment or establishments in home-health services, with a weaker wage response if labor supply expands. Because County Business Patterns measures covered establishments, the data capture formal-market adjustment rather than household employment.

The effect on Medicare is less direct. Greater home-based capacity could substitute away from inpatient and skilled-nursing care, lowering spending. It could instead complement physician and hospital care by connecting beneficiaries to the medical system, improving follow-up, or revealing unmet needs. Sanctuary policies may also affect nonlabor channels: trust in institutions, household economic security, caregiver

availability, and the location or plan enrollment of beneficiaries. Total spending can therefore increase even if the initiating shock expands noninstitutional care.

These mechanisms imply three empirical predictions. A labor-supply mechanism should produce a policy response in home-health employment before or alongside Medicare effects. A substitution mechanism should pair home-health expansion with lower institutional use. An access or complementarity mechanism can pair stronger home-based capacity with higher measured inpatient use or spending. The timing and joint pattern across outcomes are more informative than a single coefficient.

2.4 Why elder care is especially exposed

Elder care combines high labor intensity, limited substitutability, and strong preferences over setting. Assistance with bathing, dressing, meals, transportation, medication, and household tasks cannot be stored or moved offshore. When paid help becomes scarce, families supply more time, older adults go without assistance, or care shifts toward medical and institutional settings. These margins connect immigration policy to health expenditures even when Medicare does not finance most long-term custodial care.

Home-health services occupy an intermediate position. Medicare home health is a medical benefit generally tied to skilled need and homebound status, whereas much day-to-day assistance is paid privately or financed by Medicaid. Workers and agencies can nevertheless operate across adjacent markets, and household caregivers can complement formal clinical services. An enforcement shock to the broad home-care labor pool can therefore alter Medicare use without appearing one-for-one in a Medicare home-health measure.

Hospitals and skilled-nursing facilities face different adjustment costs. They require licensed staff, physical capacity, and regulatory approval, making rapid entry unlikely. Home-health agencies can adjust employment more flexibly, but certification and reimbursement still constrain supply. These differences motivate the dynamic design: employment may respond before establishments, and utilization or spending may respond only after referral networks and patient care arrangements adjust.

3 Data

3.1 Sample and policy dates

The unit of observation is the county-year, identified by a five-character Federal Information Processing Standards code. The design sample follows Hausman's set of 296 counties, chosen for their relatively large Hispanic populations and resulting policy relevance. The panel spans 2007 through 2024. Exactly half the

counties adopt during the observation window. Because this is a selected set of larger Hispanic-population counties, estimates should not be extrapolated mechanically to all U.S. counties.

Table 1. Distribution of first sanctuary-policy years

First policy year	2008	2010	2011	2012	2013	2014	2015	2017
Counties	1	1	3	2	40	86	6	9

Notes: One observation per treated county. Adoption is concentrated in 2013 and 2014, which contain 40 and 86 counties, respectively.

3.2 Care-sector outcomes

County Business Patterns (CBP) and the Quarterly Census of Employment and Wages (QCEW) supply annual employment and establishment measures for home-health care, services for elderly persons and persons with disabilities, and nursing-care facilities. CBP also supplies payroll and annual pay. We use logarithms for positive-valued counts and dollar outcomes and rates per relevant county population when levels aid interpretation. Suppression, coverage, and industry coding changes generate unbalanced samples. The independent sources permit a direct replication test: an employer-based workforce response is treated as established only when its timing and direction agree across sources.

3.3 Public ACS occupation and citizenship outcomes

We construct a separate two-period mechanism dataset from the free public ACS Equal Employment Opportunity tabulations. The 2006–2010 EEO-ALL02R, EEO-CIT02R, and EEO-NCIT02R residence tables provide county detailed-occupation counts for all, citizen, and noncitizen workers. The 2014–2018 EEO-ALL02R and EEO-CIT02R tables provide corresponding postperiod counts; noncitizen employment equals the total minus citizen count. The primary harmonized occupation combines nursing, psychiatric, and home-health aides (preperiod code 3600 and postperiod code 3601). A secondary broad measure adds personal-care employment, although the later category is less precisely harmonized.

The public detailed-occupation tables impose population thresholds. We restrict this analysis to 2013 and 2014 adopters and never-treated counties observed in both periods, producing 242 counties: 111 treated and 131 comparison counties. Citizenship outcomes contain 231 complete counties. We estimate the treated-minus-comparison change with state fixed effects, state-clustered inference, and predetermined 2010 covariates. Overlap weighting, population weighting, separate cohort estimates, California exclusion, leave-

one-state-out estimates, and within-state treatment-label permutations assess sensitivity. This medium-run repeated-cross-section exercise does not provide annual event dynamics, and citizenship does not identify undocumented status.

3.4 Medicare outcomes

County-level Centers for Medicare and Medicaid Services files provide standardized Medicare spending per beneficiary, covered inpatient and skilled-nursing stays per 1,000 beneficiaries, the share of beneficiaries using home health, and selected beneficiary characteristics. Standardized spending removes geographic payment adjustments and is logged. Utilization rates remain in their published units. The outcome universe is primarily traditional fee-for-service Medicare, so changes in Medicare Advantage participation can change the observed population even when county population is stable.

3.5 Controls and measurement

Population and socioeconomic variables come from the American Community Survey and health-resource variables from the Area Health Resources Files and the Hausman data archive. County and year fixed effects absorb time-invariant county differences and common shocks. The preferred specification does not condition on variables that could respond to the policy. We report a population-adjusted specification and assess beneficiary age, dual eligibility, risk score, Medicare Advantage participation, fee-for-service enrollment, and county population as outcomes. Missingness differs across sources; every table reports the estimation sample.

3.6 Construction of the county-year panel

All sources are merged using five-digit county FIPS codes. State and county components are stored with leading zeros before concatenation, and county names are retained only for auditing. We require uniqueness at the county-year level after each source is collapsed. Independent checks compare the set of FIPS codes across the policy file, labor files, Medicare files, and population controls. The processed panel preserves source variables alongside analysis variables so that every transformation can be reproduced.

Treatment is absorbing. For a county with effective date d , the primary treatment year is the calendar year containing d . Event time equals year minus treatment year. Never-treated counties have a missing treatment year and never enter a treated horizon. The two alternative timing rules shift late-year policies into the following year because annual employment and Medicare outcomes largely summarize activity accumulated

throughout the calendar year.

We log positive dollar and count outcomes when proportional effects are economically natural. Observations equal to zero cannot enter a log specification; suppressed employment cells are treated as missing rather than zero. Medicare stay rates and beneficiary shares remain in levels to preserve familiar units. Spending uses the standardized amount per beneficiary, which is preferable to raw reimbursement because CMS standardization removes major geographic payment adjustments.

The panel is not balanced outcome by outcome. Medicare measures are nearly complete across the policy sample, producing 4,989 observations in the baseline event study. Home-health employment produces 4,647 observations, and nursing employment 4,526. Shorter beneficiary-count and risk-score series generate only 2,072 observations in the corresponding extensions. This difference is substantive: robustness exercises using those variables compare a different period and should not be interpreted as simple reweightings of the main design.

3.7 Outcome hierarchy

Total standardized Medicare spending per beneficiary is the sole primary endpoint because it summarizes the fiscal consequence and has the most consistent temporal coverage. Inpatient-user share, inpatient covered days per 1,000 beneficiaries, and covered days per stay form the primary mechanism family. Emergency visits, emergency-user share, ambulance events, and acute readmissions form the acute-care falsification family. Skilled-nursing and home-health utilization are secondary service-setting outcomes. Employer and ACS occupation measures are exploratory mechanism tests and are interpreted only with cross-source replication and familywise multiplicity in view. This hierarchy prevents an isolated significant coefficient from carrying the paper's conclusion.

Table 2. Pretreatment descriptive statistics

Variable	Treated mean	Treated SD	Never mean	Never SD
Population	714,975.00	989,732.12	481,049.75	519,812.80
Population age 65 or older share	0.13	0.03	0.15	0.04
Poverty rate percent	14.47	5.21	13.98	5.63
Unemployment rate percent	9.69	3.63	5.33	2.66
Home-health employment	2,780.56	4,306.87	2,702.02	4,444.05
Home-health establishments	58.71	107.53	59.37	95.62
Standardized Medicare spending per beneficiary	8,324.13	1,536.36	10,244.58	1,871.49
Inpatient stays per 1,000 beneficiaries	294.66	54.58	275.25	61.56
SNF stays per 1,000 beneficiaries	70.67	21.28	63.56	19.90

Notes: Statistics use county-years before adoption for ever-treated counties and all untreated observations

for never-treated counties. Samples vary by outcome because source data are incomplete or suppressed. Values are descriptive and are not regression-adjusted.

4 Empirical strategy

4.1 Imputation difference in differences

Let Y_{ct} denote an outcome in county c and year t , and let G_c be the county's first policy year. We estimate untreated potential outcomes using observations for which $t < G_c$ and all observations in never-treated counties. The untreated-outcome model contains county and calendar-year fixed effects. For treated observations, the estimated treatment effect is the difference between the observed outcome and its imputed untreated potential outcome. Event-time estimates average these differences at $h = t - G_c$.

This estimator is appropriate because treatment is absorbing and adoption is staggered. Conventional two-way fixed-effects event studies can compare already-treated counties with later-treated counties and assign nonconvex weights when treatment effects vary over cohorts or time (Goodman-Bacon, 2021; Sun and Abraham, 2021). The imputation approach uses only untreated observations to estimate the counterfactual and then aggregates cohort-time effects transparently (Borusyak, Jaravel, and Spiess, 2024). Never-treated and not-yet-treated observations supply identification.

The baseline event window contains six leads and horizons zero through eight. Standard errors are clustered by state to allow arbitrary serial and spatial dependence among counties in the same state. We report the joint test that all displayed pretreatment coefficients equal zero. Event-study leads are diagnostic rather than proof of parallel trends: low power, anticipation, and common shocks can remain. Long-horizon estimates also place greater weight on early cohorts.

4.2 Identification

The identifying assumption is that, absent adoption, treated counties would have followed the untreated outcome process estimated from never-treated and not-yet-treated counties, conditional on county and year fixed effects. Three features deserve emphasis. First, adoption clusters in 2013 and 2014, so the design is closer to a concentrated policy wave than broadly staggered adoption. Second, policy timing may respond to local politics, immigrant settlement, or litigation that also predicts health-market trends. Third, state policy and health-market shocks may affect several counties together. State-clustered inference addresses correlation in errors but does not remove state-specific confounding.

We probe these concerns by inspecting six pretreatment coefficients; shortening the event window; adding log population; excluding the pandemic years; shifting late-year adoptions forward; excluding California; and estimating beneficiary-composition outcomes. The timing shifts are especially important because annual data assign a full treatment year to policies that may have been active for only weeks. The California exclusion addresses the concentration of adopters and the possibility that state-specific health and immigration institutions drive the result.

4.3 Multiple outcomes and interpretation

The analysis uses a fixed hierarchy for interpretation. Total standardized Medicare spending per beneficiary is the primary outcome. Inpatient participation, covered days, and days per stay form the primary mechanism family because they identify extensive and intensive inpatient margins. Emergency use, ambulance events, and readmissions are falsification outcomes within the broader seven-outcome acute-access family. Employment, establishments, payroll, wages, provider capacity, and ACS occupation measures are exploratory mechanism outcomes. We report exact p values and Benjamini–Hochberg adjustments within the acute-access and ACS families. Statistical significance does not resolve compositional or identifying threats, and exploratory results are not treated as mechanisms without cross-source confirmation.

4.4 Estimand and weighting

The event-time coefficient is an average treatment effect on treated county-years observed at a given horizon. Counties receive equal weight in the unweighted baseline, subject to the estimator’s support conditions. The parameter therefore describes the average county in the estimation sample, not the average Medicare beneficiary or the national fiscal effect. This distinction is important because treated counties differ greatly in population. A beneficiary-weighted estimand would answer a different question and would place substantial weight on a small number of metropolitan counties.

Treatment-effect heterogeneity can arise across cohorts, states, baseline immigrant shares, urban scale, or the content of the local rule. The imputation estimator remains interpretable under heterogeneous effects, but the composition of the event-time average changes at long horizons as later cohorts cease to contribute. Year-eight estimates disproportionately represent early adopters. We report raw treated-county support in the appendix and avoid interpreting differences between adjacent horizons as pure within-county dynamics.

Our standard errors cluster at the state level because counties within a state share legal regimes, labor markets, Medicaid policies, hospital systems, and political shocks. With a moderate number of state clusters,

conventional cluster-robust inference can still be sensitive to influential states. The California exclusion is thus both a substantive generalizability check and an influence diagnostic. A wild-cluster bootstrap would be a useful additional inference exercise when the final state-policy specification is fixed.

4.5 Threats to identification

Policy endogeneity is the first threat. Counties that adopt sanctuary rules may be responding to demographic change, litigation, elections, or administrative reforms. County fixed effects remove permanent differences, and year fixed effects remove common shocks, but neither removes differential trends. Event-study leads provide a direct diagnostic. They are reassuring for total Medicare spending and inpatient stays and unfavorable for several labor and composition outcomes.

Concurrent policy is the second threat. State Medicaid expansions, minimum-wage laws, paid-leave rules, hospital closures, and statewide immigration policies can affect care markets around the same time. State-clustered standard errors do not solve omitted-variable bias. The concentration of adoption in 2013 and 2014 raises this concern because it coincides with Affordable Care Act implementation. Medicare eligibility and benefits were not expanded by that law in the same way as Medicaid, but delivery systems and local labor demand changed.

Measurement is the third threat. Policy dates are exact, whereas outcomes are annual. A policy adopted in October is assigned nearly the same event time as one adopted in January. The late-year timing shifts address this mismatch. Employment suppression can also be nonrandom in small county-industry cells. Treating suppressed cells as zero would manufacture declines, so we retain them as missing and interpret the labor sample as the observed formal sector.

Interference is a fourth threat. Workers, patients, and providers can cross county boundaries. A protective policy in one county may attract labor from neighboring counties or serve residents from a larger referral region. Such spillovers violate the no-interference condition and can attenuate or redistribute estimates, especially when neighboring counties adopt together. County-level effects should therefore be read as local equilibrium responses within administratively defined markets rather than isolated biological treatment effects.

5 Policy First Stage: The Local Jail-to-ICE Pipeline

The Medicare analysis is meaningful only if the recorded policy dates correspond to a material change in local enforcement practice. A legal label by itself may be weak treatment: counties may announce a policy without implementing it, comply through informal channels, or face federal substitution that leaves realized enforcement unchanged. This section therefore reconstructs the administrative sequence connecting an ICE request to federal custody. The purpose is treatment validation. Hausman (2020) already establishes that sanctuary policies reduced detainers, detainer-based arrests, and local-fingerprint removals. The analysis here asks more narrowly which link in the request-to-custody pipeline changes in the counties and years available for administrative reconstruction.

5.1 Administrative records and outcome definitions

The raw detainer file contains 1,799,842 individual ICE records. Facility identifiers are mapped to counties using Hausman’s manually constructed crosswalk. The annual estimation panel contains 113 counties in 31 states observed from 2007 through 2017, including the large 2013 and 2014 adoption cohorts and counties that remain untreated during the observation window. To reduce contamination of the comparison group, the analysis removes five counties with verified post-2017 changes in local cooperation and two Illinois counties affected by the 2017 TRUST Act.

The administrative disposition field permits a decomposition of the pipeline. A custody success is a detainer ending in “Booked into Detention” or “Transferred.” The custody-success rate divides these dispositions by all detainers. The recorded-decline rate divides records coded “Detainer Declined by LEA” by all detainers. The notification-request rate measures whether ICE requested notice before release. It is important that the last variable records the federal request, not whether a county failed to notify ICE. The count analysis also examines total detainers, custody successes, and recorded declines.

The first-stage design follows the main chapter’s imputation approach. It includes county and year fixed effects, event years zero through three, six pretreatment coefficients, and standard errors clustered by state. Reported post-policy averages give equal weight to event years zero through three. The shorter postperiod reflects the administrative file’s endpoint and avoids treating missing later years as zeros.

Table FS1. Sanctuary adoption and the ICE detainer pipeline

Outcome	Average effect, years 0–3	<i>p</i> value	Pretrend <i>p</i>
Log detainees issued	-0.344	0.023	0.022
Log ICE custody successes	-0.884	< 0.001	0.644
Custody-success rate	-0.278	< 0.001	0.546
Log recorded declines	0.873	< 0.001	< 0.001
Recorded-decline rate	0.108	< 0.001	0.104
Notification-request rate	0.010	0.655	0.098

Notes: Count outcomes are in logs. Rate effects are proportions; -0.278 equals a 27.8-percentage-point decline. Estimates use *did_imputation*, county and year fixed effects, and state-clustered standard errors. The pretrend column jointly tests six lead coefficients.

5.2 First-stage results

The conditional custody margin changes sharply. Averaged over years zero through three, the custody-success rate falls by 27.8 percentage points. The adoption-year coefficient is a smaller and imprecise decline of 7.5 points, followed by declines of 31.3, 32.0, and 40.5 points in years one through three. The six pretreatment coefficients are jointly insignificant. In parallel, the recorded-decline rate rises by 10.8 percentage points on average, with a joint pretreatment *p* value of 0.104. These rate outcomes indicate that sanctuary adoption changes what happens after ICE lodges a request.

ICE's recorded demand for information does not respond comparably. The notification-request rate changes by only one percentage point and is statistically indistinguishable from zero. This distinction is institutionally informative: the administrative break occurs in the local response and subsequent transfer to federal custody, not in an immediate disappearance of ICE requests.

The count of detainees falls by 0.344 log points, approximately 29 percent after exponentiation. Its joint pretreatment test rejects, however, so it cannot carry a clean causal claim. The log count of custody successes falls by 0.884 log points, approximately 59 percent, with a favorable pretreatment diagnostic. The count and rate evidence together imply a substantial reduction in the realized effectiveness of detainees, while the evidence that ICE reduced its initial issuance of detainees is weaker.

5.3 Robustness and interpretation

California does not account for the conditional-rate findings. Excluding California produces a 28.7-percentage-point decline in custody success and a 10.1-point increase in recorded declines. Adding state-specific linear trends attenuates the custody estimate to -12.6 points (*p* = 0.057) and the decline estimate to 7.2 points (*p* = 0.013). The direction therefore survives, although the estimated magnitude depends on how differential state trajectories are modeled. State-by-year fixed effects cannot identify the effect in this

Conditional enforcement margins

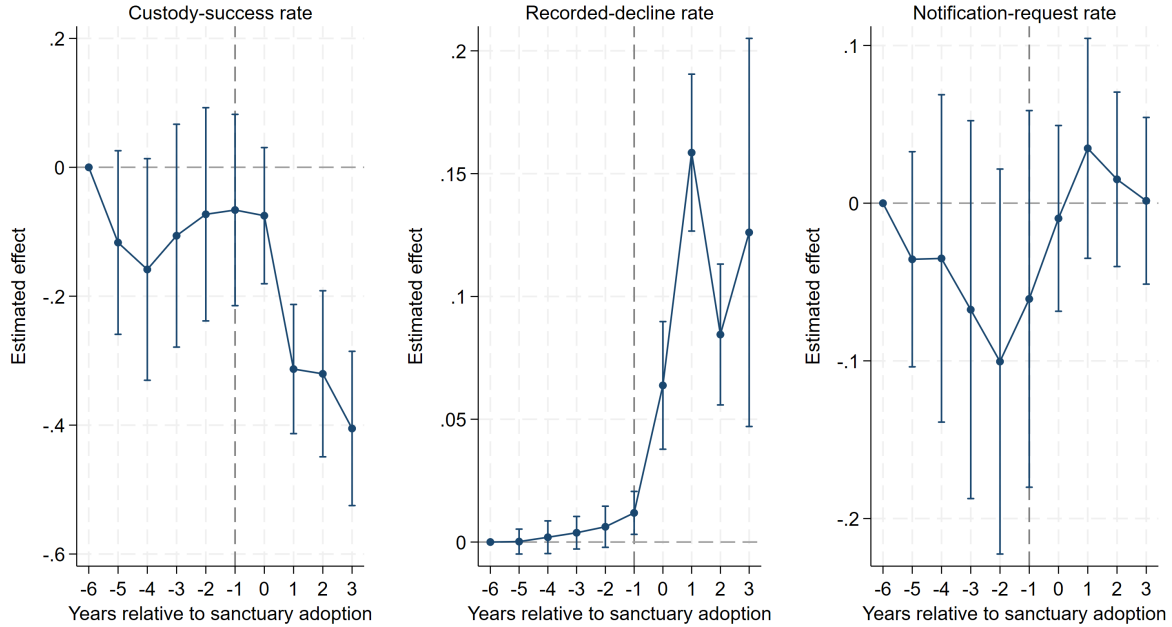


Figure FS1. Sanctuary adoption and conditional ICE-enforcement outcomes

Notes: Points are imputation difference-in-differences estimates; shaded regions are 95 percent confidence intervals using state-clustered standard errors. The notification outcome records an ICE request for release notice, not a notification failure.

restricted administrative sample because policy timing supplies essentially no usable within-state variation.

These estimates validate the policy treatment but do not establish the downstream healthcare mechanism. They show that adoption changed local cooperation in a manner large enough to alter realized federal custody. They do not show whether ICE substituted community arrests for jail arrests, whether affected individuals were later detained elsewhere, or which criminal-history groups account for the decline. Nor do they imply that the people whose custody outcomes changed were Medicare beneficiaries. The first stage establishes a meaningful institutional shock; the remaining sections ask whether county Medicare outcomes also change after that shock.

6 Medicare Spending and Utilization Results

6.1 Medicare spending and inpatient utilization

Figure 1 shows no detectable change in standardized Medicare spending during the first two post-policy years. The effect becomes positive in year three and grows thereafter. Table 3 reports selected horizons. At year five, the log-point coefficient is 0.0222 (SE 0.0068; $p = 0.001$), equivalent to 2.24 percent. At year eight, it

is 0.0442 (SE 0.0108; $p < 0.001$), equivalent to 4.52 percent. The joint pretreatment p-value is 0.183.

The economic magnitude is nontrivial. Among the 133 treated counties with valid baseline spending and enough calendar support to contribute at year eight, beneficiary-weighted standardized spending in the year before adoption averages approximately \$9,457. Applying the pooled 4.52 percent estimate corresponds to about \$428 per fee-for-service beneficiary. Applied mechanically to their 8.12 million baseline fee-for-service beneficiaries, this is approximately \$3.47 billion. This calculation translates the coefficient into baseline dollars; it is not a national budget estimate, does not incorporate enrollment changes, and should not be interpreted causally beyond the identifying assumptions of the pooled design.

Inpatient covered stays display a similarly delayed profile. The estimates are positive but imprecise through year five. They reach 11.3 additional stays per 1,000 beneficiaries in year seven and 14.0 in year eight. The six leads are jointly insignificant ($p = 0.771$). These results are consistent with a delayed increase in intensive health-care use, but aggregate utilization cannot reveal whether admissions are avoidable, reflect improved access, or follow changes in beneficiary health.

The shapes of the two event studies reinforce one another. Spending turns positive in year three, while inpatient stays rise gradually and become precise later. Standardized inpatient spending also increases at long horizons, although its six-lead pretrend test is borderline rather than fully reassuring. Skilled-nursing and home-health spending components do not deliver clean complementary evidence: skilled-nursing spending is imprecise, and home-health spending has strong differential pretrends. Total spending is therefore more credible than a decomposition into mutually exclusive service channels.

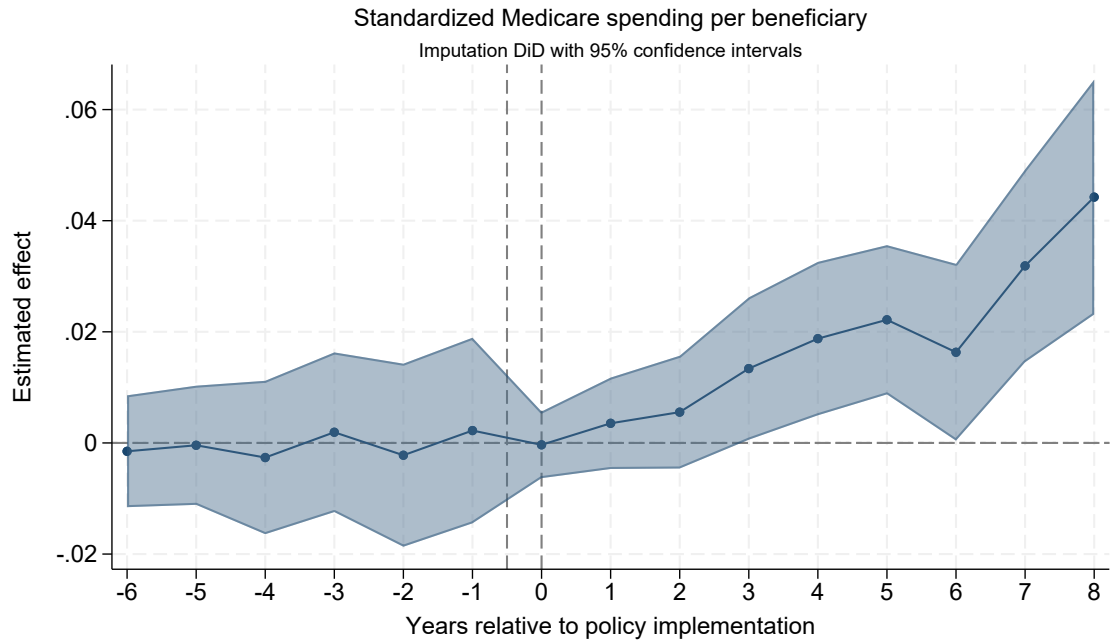


Figure 1. Sanctuary policies and standardized Medicare spending per beneficiary. Notes: Points are imputation difference-in-differences estimates relative to untreated potential outcomes. Bars show 95 percent confidence intervals with standard errors clustered by state.

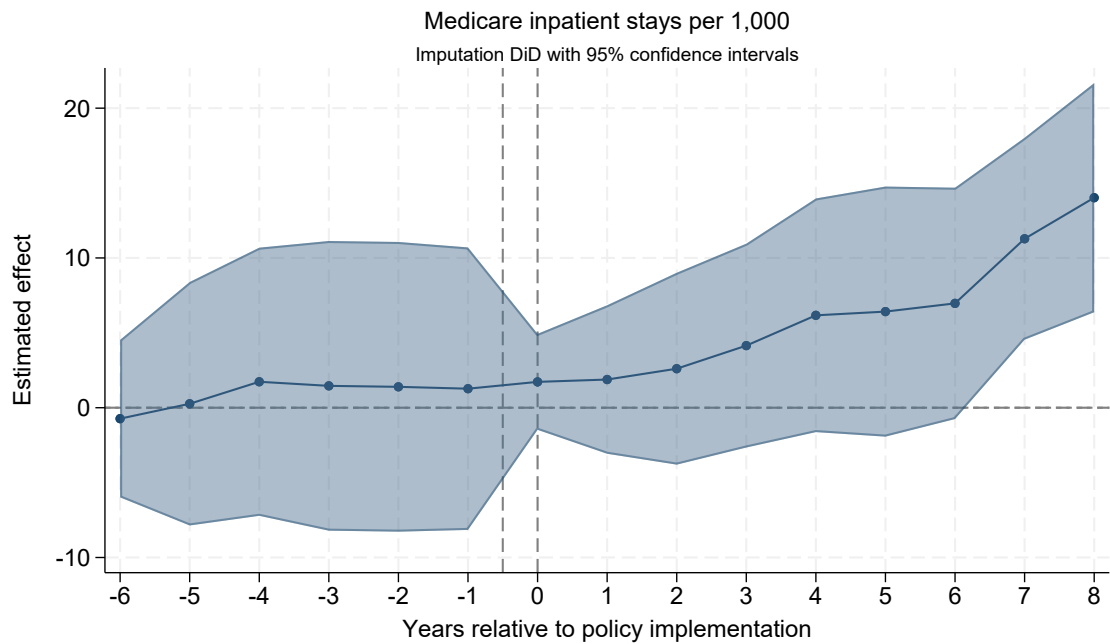


Figure 2. Sanctuary policies and inpatient covered stays per 1,000 beneficiaries. Notes: Points are imputation difference-in-differences estimates. Bars show 95 percent confidence intervals with standard errors clustered by state.

Table 3. Dynamic effects on Medicare outcomes

Event time	Spending percent	Inpatient stays	SNF stays	Home-health share
Year 0	-0.03 (0.31)	1.73 (1.63)	-0.96 (0.69)	0.00 (0.00)
Year 3	1.35** (0.65)	4.15 (3.47)	-0.88 (1.67)	0.01* (0.00)
Year 5	2.24*** (0.68)	6.42 (4.26)	-0.89 (2.12)	0.01* (0.01)
Year 8	4.52*** (1.08)	14.02*** (3.90)	0.95 (3.72)	0.01* (0.01)
Pretreatment p	0.183	0.771	0.682	0.010
Observations	4,989	4,989	4,989	4,989

Notes: Spending entries transform log coefficients to percentages; their reported standard errors use the log-point approximation multiplied by 100. Other entries are in published units: stays per 1,000 beneficiaries and the home-health beneficiary share. Parentheses contain standard errors clustered by state. The pretreatment row reports the joint test for event times -6 through -1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The formal care-sector event studies and Callaway–Li quantile treatment effects are reported in Supplementary Sections S1 and S3. Their inclusion in the supplement preserves the full prespecified analysis while keeping the article focused on the primary spending outcome and its inpatient decomposition.

6.2 Robustness and treatment timing

Table 4 focuses on the primary spending outcome. Population adjustment and pandemic-year exclusion leave the year-five estimate close to 2.5 percent. Coding July-through-December policies as beginning the next year or shifting only fourth-quarter adoptions produces nearly identical dynamics and improves the joint pretreatment p-value. These exercises reduce concern that partial-year treatment mechanically generates the delayed pattern.

California is the consequential exception. Removing its counties lowers the year-five effect to 1.66 log points (1.66 percent; $p = 0.057$) and the year-eight effect to 2.65 log points (2.69 percent; $p = 0.050$). The pretreatment p-value also falls to 0.088. The direction survives, but precision and magnitude weaken. The national interpretation therefore depends partly on California, whose immigration and health-policy environment is distinctive.

The alternative timing estimates are informative beyond statistical significance. Under the July shift, the spending effect is 1.6 log points in year three, 2.3 in year five, and 4.8 in year eight. Under the October shift, the corresponding estimates are 1.5, 2.4, and 4.8 log points. The similarity indicates that late-year adopters do not create the baseline pattern merely because their nominal treatment year contains little actual exposure.

Excluding 2020 and 2021 addresses the sharp, geographically uneven disruption in utilization during the COVID-19 pandemic. The year-five estimate remains 2.5 percent, close to the baseline. This does not remove

all pandemic influence from later horizons, because post-pandemic years may contain delayed utilization and mortality selection. It shows, however, that the main medium-horizon estimate is not driven solely by the two most disrupted calendar years.

Table 4. Robustness of standardized Medicare spending

Specification	Year 5 percent	Year 8 percent	Pretreatment p	Observations
Baseline	2.24*** (0.68)	4.52*** (1.08)	0.183	4,989
Four-lead window	2.24*** (0.68)	—	0.301	4,558
Population adjusted	2.39*** (0.70)	—	0.314	4,558
Exclude 2020 and 2021	2.52*** (0.67)	—	0.301	4,238
Shift July to December	2.30*** (0.81)	4.91*** (1.49)	0.255	5,074
Shift October to December	2.44*** (0.88)	4.97*** (1.65)	0.356	5,045
Exclude California	1.66* (0.87)	2.69** (1.35)	0.088	4,436

Notes: Estimates are percentage changes from log specifications. Parentheses contain standard errors using the log-point approximation multiplied by 100. The four-lead and population-adjusted models are estimated only through year five. Late-year timing specifications assign affected adoptions to the following calendar year. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.3 Beneficiary composition

Composition tests reported in Supplementary Table S2 reveal a central limitation. Average beneficiary age does not change differentially, and the available risk-score series has no joint pretrend rejection, although its post-treatment support is too short to identify the requested long horizons. By contrast, dual eligibility rises and Medicare Advantage participation falls in treated counties, but both patterns begin before adoption. Fee-for-service enrollment and county population also show differential pretrends. These findings do not mechanically invalidate standardized per-beneficiary spending, but they show that the observed Medicare population evolves differently in treated counties.

Because the Medicare spending series pertains to beneficiaries observed in fee-for-service data, differential movement into or out of Medicare Advantage can alter who remains in the denominator. A beneficiary-weighted specification is not informative in the current extract: the beneficiary-count series covers only a shorter period and the resulting model has a severe pretrend rejection. A stronger final design would harmonize enrollment counts over the full analysis window or use beneficiary-level claims and enrollment data to hold the population definition constant.

6.4 What the null results rule out

Null and imprecise findings are useful when interpreted at the scale supported by their confidence intervals. The skilled-nursing-stay estimates exclude a large, immediate increase comparable to the inpatient pattern, but they allow moderate changes at long horizons. Establishment and wage estimates are substantially less precise and cannot rule out economically relevant responses. The absence of statistical significance should not be translated into evidence of exact zero effects.

The joint outcome pattern does rule against the simplest version of the hypothesized mechanism. If sanctuary policies produced a discrete expansion of formal home-health capacity that promptly substituted for institutional care, one would expect clean increases in employment and establishments, greater home-health use, and declines in inpatient or skilled-nursing use. We do not observe that combination. The delayed spending increase requires a broader mechanism or a compositional explanation.

The findings also rule against an immediate Medicare billing response to policy coding. Event-time-zero spending is essentially zero, and the first two horizons are small. A contemporaneous coding change, one-time outreach campaign, or mechanical annual alignment would be more likely to generate a discontinuity at adoption. The gradual profile shifts attention toward cumulative exposure and evolving county composition.

6.5 Cohort decomposition and jurisdiction influence audit

The baseline imputation estimates cluster standard errors at the state level because counties within a state can share legislation, enforcement institutions, and health-market shocks. We re-estimate the complete model after omitting each of the 26 treated states and each of the 148 treated counties; Supplementary Table S4 and Figure S4 report the complete diagnostics. We also isolate the two large adoption cohorts and compare each exclusively with never-treated counties.

No individual jurisdiction accounts for the delayed spending result. Across all 26 treated-state exclusions, year-8 estimates remain positive and statistically significant, ranging from 0.027 when California is omitted to 0.051 when Florida is omitted. At year 5, the range is 0.016 to 0.026; 25 of 26 estimates are significant at five percent. Omitting each treated county produces a narrower year-8 range of 0.043 to 0.047; all 148 estimates remain positive and significant. The corresponding year-5 estimates range from 0.021 to 0.024 and are all significant.

California remains quantitatively influential: excluding it lowers the year-8 estimate from 0.044 to 0.027 log points and leaves a borderline p value of 0.050. This is attenuation rather than disappearance, and no single California county reproduces the state-level change. The evidence is inconsistent with a single-county

outlier but remains partly sensitive to the common contribution of California counties.

The cohort decomposition is less reassuring about causal interpretation. Year-8 estimates are 0.075 log points for 2013 adopters and 0.037 for 2014 adopters, but joint tests reject the six placebo coefficients for both cohorts ($p < 0.001$ and $p = 0.005$). The positive long-run pattern appears in both large adoption groups, but the cohort estimates do not independently validate parallel trends.

These diagnostics strengthen the conclusion that the pooled result is not generated by a single treated jurisdiction. They do not resolve the cohort-specific pretrend failures or rule out shocks shared across adopting counties.

6.6 Comparability-adjusted risk-set estimates

The pooled event study is stable to individual-jurisdiction exclusions, but stability is distinct from comparability. We therefore construct separate risk sets for the 2013 and 2014 adoption cohorts, each compared exclusively with never-treated counties. The balancing vector includes six years of standardized-spending levels, the fitted prepolicy spending trend, baseline population, Medicare Advantage participation, beneficiary risk, inpatient use, nursing-bed capacity, and predetermined socioeconomic and hospital-capacity measures. Exact entropy balance is infeasible for 2013, so the primary specification uses regularized entropy weights and reports residual balance and effective sample sizes. The full algorithm, tuning grid, balance plots, overlap distributions, and event paths appear in the supplementary appendix.

Table 5. Comparability-adjusted spending estimates

Cohort	Specification	$h = 5$	$h = 8$	Pretrend p
2013	Regularized entropy weights	0.007 (0.013)	0.047 (0.019)	< 0.001
2013	Entropy weights, excluding California	-0.063 (0.031)	-0.062 (0.042)	0.238
2014	Regularized entropy weights	0.019 (0.011)	0.022 (0.017)	0.829
2014	Weights capped at p95	0.024 (0.010)	0.030 (0.015)	0.526

Notes: State-clustered standard errors are in parentheses. Each cohort is compared only with never-treated counties. Year -1 is omitted; the pretrend column reports a joint test of years -6 through -2 .

The 2014 design produces the strongest comparability-adjusted evidence. Weighting reduces the largest absolute standardized mean difference from 0.681 to 0.013, all included covariates fall below 0.10, and the effective comparison group contains 55 counties. The primary year-eight coefficient is 0.022 log points and retains the delayed positive pattern, although it is imprecise ($p = 0.195$). Capping weights at their 95th percentile yields 0.030 log points ($p = 0.048$), with a clean pretrend test. The evidence supports the direction and timing of the pooled result for 2014, while precision depends on the weighting rule.

The 2013 design does not supply a credible matched counterfactual. Its effective comparison group falls

to 21.5 counties, its maximum absolute standardized difference remains 0.281, and its weighted pretrend test rejects. Excluding California removes the pretrend rejection but reverses the year-eight estimate to -0.062 log points. We therefore do not treat 2013 as independent causal confirmation. The pooled estimate summarizes a stable reduced-form association across treated counties; the 2014 cohort provides the cleanest observables-adjusted corroboration.

6.7 Direct evidence on access and care intensity

We use previously unexploited fields in the Medicare Geographic Variation files to distinguish the extensive margin of inpatient use, treatment intensity conditional on an admission, and emergency-care pathways. All models use the primary imputation estimator, policy timing, six placebo coefficients, and state-clustered inference. We control the false discovery rate within the prespecified seven-outcome acute-access family at each event horizon.

Table 6. Medicare access and acute-care decomposition

Outcome	$h = 5$	$h = 8$	Adjusted q , $h = 8$	Pretrend p
Beneficiaries with inpatient use	0.224 (0.207)	0.530 (0.181)	0.012	0.923
Inpatient covered days per 1,000	69.957 (29.174)	125.839 (34.361)	0.002	0.645
Inpatient covered days per stay	0.138 (0.069)	0.263 (0.112)	0.045	0.481
Acute readmission rate	0.087 (0.221)	0.198 (0.185)	0.399	0.171
Emergency visits per 1,000	6.066 (15.115)	22.966 (14.042)	0.178	0.802
Beneficiaries with emergency use	-0.050 (0.430)	0.390 (0.476)	0.481	0.774
Ambulance events per 1,000	-6.445 (19.667)	8.458 (17.733)	0.633	0.729

Notes: State-clustered standard errors are in parentheses. Beneficiary-share outcomes are reported in percentage points. Adjusted q values use the Benjamini–Hochberg procedure across the seven acute-access outcomes at the stated horizon. The pretrend column reports the joint test of six placebo coefficients.

At year 8, the share of beneficiaries with inpatient use rises by 0.530 percentage points, covered inpatient days rise by 125.8 per 1,000 beneficiaries, and covered days per stay rise by 0.263. Their adjusted q values are 0.012, 0.002, and 0.045, respectively, and their joint pretrend p values range from 0.481 to 0.923. The spending increase therefore coincides with both a wider inpatient-user margin and greater treatment intensity among stays.

Emergency visits, the share using emergency care, ambulance events, and acute readmissions remain statistically indistinguishable from zero and have clean pretreatment profiles. These nulls do not prove that additional admissions are elective or welfare improving, but they weigh against a broad increase in emergency episodes, ambulance dependence, or rapid rehospitalization as the explanation for higher spending.

6.8 Outpatient access and service composition

The preceding decomposition identifies inpatient utilization as the proximate spending margin but does not establish why inpatient use rises. A natural hypothesis is that a less threatening enforcement environment increases engagement with routine care, which could reveal previously untreated conditions and generate referrals for hospital treatment. To evaluate that pathway, we recover additional county fields from the public Medicare Geographic Variation files: outpatient visits and users, evaluation-and-management events and users, FQHC/RHC visits and users, procedures, tests, and imaging. Legacy percentage fields reported on a zero-to-100 scale are harmonized to the zero-to-one scale used beginning in 2014. Coverage is complete or nearly complete for the 296 counties from 2007 through 2024.

Table 7. Medicare outpatient-access mechanism tests

Outcome	Years 0–3	<i>p</i>	Years 4–8	<i>p</i>	Pretrend <i>p</i>
Outpatient visits per 1,000	-102.4	0.087	-105.6	0.135	0.468
Beneficiaries using outpatient care	-0.0070	0.015	-0.0041	0.409	0.667
Evaluation-and-management events per 1,000	-325.6	0.004	-256.6	0.112	0.152
Beneficiaries using evaluation and management	0.0018	0.140	0.0033	0.265	0.118
FQHC/RHC visits per 1,000	36.8	< 0.001	88.9	< 0.001	< 0.001
Beneficiaries using FQHC/RHC care	0.0085	< 0.001	0.0162	0.001	0.025
Test events per 1,000	506.6	0.050	717.3	0.035	0.118
Beneficiaries receiving tests	0.0063	0.017	0.0123	0.001	0.124
Imaging events per 1,000	138.4	< 0.001	232.4	< 0.001	< 0.001

Notes: Entries are equal-weighted averages of event-time coefficients. Share outcomes are proportions; 0.007 equals 0.7 percentage points. Models use the primary imputation estimator and state-clustered inference. Benjamini–Hochberg adjustments across the twelve tested access outcomes are reported in the replication files.

The access results do not form a coherent bridge to the delayed inpatient response. In years zero through three, outpatient participation falls by 0.70 percentage points and evaluation-and-management events fall by 326 per 1,000 beneficiaries. Both survive false-discovery-rate adjustment across the twelve access outcomes, but their graphs show gradual downward movement before adoption even though the joint lead tests do not reject. The estimates also occur immediately and dissipate at longer horizons, unlike the gradual Medicare-spending profile.

Testing moves in the opposite direction. The share of beneficiaries receiving tests rises by 0.63 percentage points in years zero through three and by 1.23 points in years four through eight. The formal pretreatment test is acceptable, but the individual leads trace an upward slope. FQHC/RHC and imaging measures increase strongly, yet their joint pretreatment tests reject. These estimates therefore cannot be used as causal mechanisms. Procedure outcomes are similarly sensitive to pretreatment movement and do not clarify the pattern.

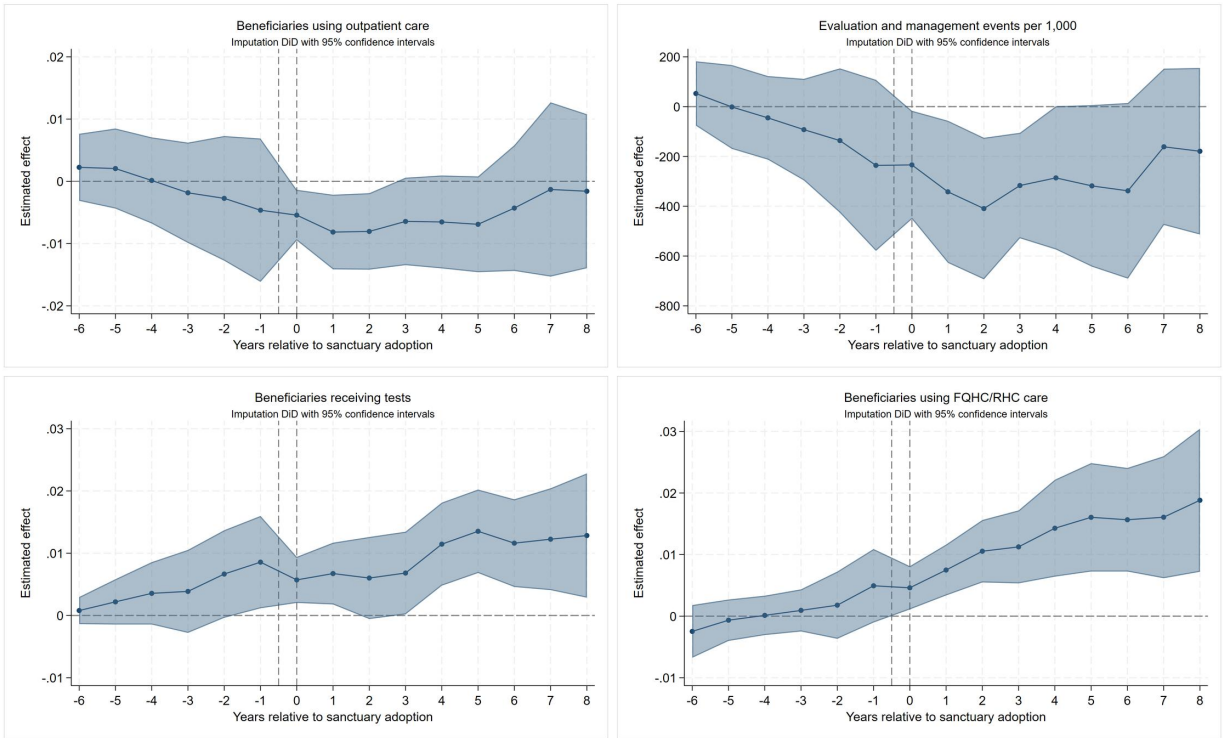


Figure 3. Medicare outpatient-access mechanism tests

Notes: Each panel reports imputation difference-in-differences estimates and 95 percent confidence intervals. The upper panels have acceptable formal joint lead tests but visible pre-adoption movement. The FQHC/RHC panel rejects the joint pretreatment test.

The combined pattern may represent a change in service composition—fewer broad outpatient contacts but greater testing among those who engage with care—or it may reflect coding, provider practice, or beneficiary selection. It does not support the simpler claim that sanctuary adoption broadly increases routine outpatient access and thereby generates later inpatient admissions. Public CMS files contain Prevention Quality Indicator fields, but those fields are populated only for state and national rows and are blank for counties in every available year. The chapter therefore cannot test county ambulatory-care-sensitive admissions using this public product.

6.9 Direct-care workforce evidence and cross-source validation

QCEW employment in services for elderly persons and persons with disabilities (NAICS 624120) increases sharply after adoption. At year 8, the estimate is 4.10 additional workers per 1,000 residents, with a joint pretrend p value of 0.138. QCEW establishment counts also increase. Matched County Business Patterns measures do not replicate either result: the year-8 employment-rate estimate is 0.49 ($p = 0.276$), and the establishment estimate is negative and imprecise. The QCEW effect is unusually large at adoption and substantially larger for the 2013 cohort than for the 2014 cohort. Although it survives individual state exclusions, some exclusion-specific pretrend tests fail.

Because two federal employer datasets disagree on the same industry, we do not interpret the QCEW pattern as established evidence that sanctuary policies expand the direct-care workforce. Differences in disclosure suppression, coverage, and annual reporting may generate source-specific changes. The public ACS EEO tabulations provide an independent occupation-based test and directly separate citizen from noncitizen workers.

Table 8. Public ACS direct-care workforce mechanism test

Outcome	Estimate	Cluster SE	p	BH q
Log core aides	0.0878	0.0353	0.017	0.172
Core aides per 1,000 population	0.7697	0.4963	0.129	0.644
Core aides per 1,000 age 65+	3.1064	3.5696	0.389	0.738
Log noncitizen core aides	0.1674	0.1825	0.365	0.738
Noncitizen share of core aides	0.0095	0.0136	0.490	0.738

Notes: Each coefficient is the treated-minus-comparison difference between the 2006–2010 and 2014–2018 EEO tabulations. The sample contains 2013/2014 adopters and never-treated counties observed in both periods. Models include state fixed effects and 2010 covariates; standard errors are clustered by state. Benjamini–Hochberg q values adjust across ten ACS occupation outcomes. Core aides combine nursing, psychiatric, and home-health aides.

The total number of core aides rises by 0.088 log points in the covariate-adjusted model. Overlap weighting produces a similar estimate of 0.091, excluding California yields 0.086, and the within-state permutation

p value is 0.052. The result is nevertheless incomplete as mechanism evidence. Its ten-outcome q value is 0.172, population weighting reduces precision, and the effect is concentrated in the 2014 cohort: the 2013 estimate is -0.003 while the 2014 estimate is 0.089. More consequentially, neither the aide rate per older resident nor the noncitizen count or share differs significantly. The ACS evidence therefore does not show that treated counties gained greater direct-care capacity through noncitizen labor.

Across QCEW, CBP, and ACS, the workforce evidence is source-sensitive and fails the paper’s cross-source confirmation rule. The cleaner mechanism evidence lies in the Medicare decomposition: more beneficiaries use inpatient care and covered stays become longer, without corresponding increases in emergency use or readmissions. We consequently treat an immigrant labor-supply channel as unresolved rather than as the explanation for higher spending.

7 Interpretation and mechanisms

The most defensible empirical statement is that sanctuary-policy adoption is followed by a delayed rise in standardized Medicare spending and inpatient utilization in this selected county sample. The pooled spending outcome is measured per beneficiary and standardized for geographic payment differences, its pretreatment coefficients are jointly insignificant, and the pattern survives conservative annual timing rules, every treated-state exclusion, and every treated-county exclusion. Inpatient-user share, covered days, and days per stay show mutually consistent long-run changes, clean pretreatment profiles, and multiplicity-adjusted significance. Cohort-specific spending pretrend tests reject, however, so we treat the pooled spending result as the primary reduced-form evidence and the cohort decomposition as a limitation rather than independent validation.

The decomposition in Table 6 identifies the proximate spending margin, although it does not identify the underlying behavioral mechanism. The administrative first stage establishes that the legal treatment materially reduced the conversion of detainees into ICE custody, but this is evidence of implementation rather than mediation. The outpatient analysis further shows that broad routine-care engagement does not supply the missing bridge. More beneficiaries receive inpatient care and covered days per stay increase, while emergency visits, ambulance events, and readmissions do not. The workforce tests provide no consistent bridge from policy to utilization: QCEW is positive, CBP does not replicate it, and public ACS occupation data show no robust change in direct-care capacity or noncitizen workforce composition. The evidence therefore supports an inpatient-utilization channel but not workforce mediation.

One possible interpretation is improved connection to formal care. Reduced enforcement salience may

stabilize mixed-status households and caregiver networks or increase willingness to interact with local institutions. Greater contact can raise recorded spending when previously unmet needs are treated. A second interpretation is delayed presentation or greater severity, which could generate both a wider inpatient-user margin and longer stays. Provider admission practices, referral behavior, and coding are also observationally compatible. Finally, declining Medicare Advantage participation or rising dual eligibility may change the fee-for-service population. Pretreatment movement in those composition measures prevents a clean treatment interpretation but leaves selection as a competing explanation.

The lag is also informative. An immediate labor-supply or coding response should appear near adoption. Instead, spending and inpatient use grow over several years. Slow provider entry, cumulative health effects, aging of exposed households, and cohort composition can all generate such a profile. The current county aggregates cannot distinguish them. Claims that higher spending represents worse health, better access, or inefficiency would therefore exceed the evidence.

7.1 Access and complementarity

Higher spending need not imply reduced welfare. When beneficiaries face unmet need, an increase in utilization can represent improved access to effective care. Caregivers often identify symptoms, organize transportation, reconcile medications, and communicate with clinicians. A more stable caregiving environment can therefore increase appropriate admissions and post-acute follow-up. County aggregates do not measure quality, avoidable hospitalization, mortality, or patient-reported access, so the welfare sign remains open.

Complementarity remains possible when household assistance enables covered medical care. Medicare generally does not pay for long-term custodial support, but it pays for hospital, physician, post-acute, and qualifying home-health services. The ACS null results indicate that any such pathway is not expressed as a robust county-level increase in noncitizen core aides or aides per older resident. Informal caregiving, worker hours, and within-household behavior remain unmeasured.

A competing pathway operates through institutional trust. Sanctuary policies may signal that local institutions are less connected to immigration enforcement. Although Medicare beneficiaries themselves are generally citizens or lawfully present, they may live in mixed-status households or rely on immigrant relatives. Reduced fear could change interaction with providers. This interpretation remains untested because the county files do not observe household immigration exposure, admission source, preventive care, language access, or the beneficiary's connection to immigrant family members.

7.2 Health, substitution, and induced demand

A health channel could move spending in either direction. Lower household stress and greater caregiver stability may improve chronic-disease management and eventually reduce acute use. Better detection can initially raise spending as previously unmet needs are treated. If the effect accumulates through chronic conditions, a delayed response is plausible. The current analysis lacks mortality, diagnosis-specific admissions, and ambulatory-care-sensitive hospitalization measures needed to distinguish detection from deterioration.

Provider behavior offers another explanation. Where labor constraints ease, agencies and health systems may accept more patients, increase service intensity, or cultivate referrals. Such supply-induced use would raise spending but should eventually appear in capacity or employment. The weak establishment response and uncertain employment design provide limited support. Provider-level Medicare enrollment and claims would allow decomposition into the number of providers, beneficiaries per provider, services per beneficiary, and payment per service.

Substitution away from skilled nursing receives little support. Skilled-nursing covered stays remain near zero, while inpatient stays rise. Yet Medicare SNF stays capture short-term post-acute care rather than long-term nursing-home residence, the outcome most closely related to aging in place. The null SNF result therefore does not directly contradict evidence that immigration enforcement changes institutional residence. A valid test requires nursing-home census, minimum data set, or American Community Survey group-quarters measures.

7.3 Composition as an alternative explanation

Changes in the observed beneficiary population can produce a spending effect without changing care for a fixed group. If relatively healthy beneficiaries enter Medicare Advantage more quickly in comparison counties, average fee-for-service spending can rise in treated counties. If dual-eligible beneficiaries become more prevalent, spending can also rise because this group has greater morbidity and service use. Standardization adjusts payment rates, not selection into the analytic population.

The composition event studies show strong differential movement before policy adoption, so they cannot be interpreted as clean treatment effects. Their value is diagnostic: they demonstrate that treated and comparison counties were already on different Medicare enrollment paths. Including post-treatment composition as a regression control could introduce bias because plan choice and dual status may respond to the policy or related shocks. The preferred solution is design-based: construct outcomes for a stable enrollment cohort, model plan transitions directly, or use total Medicare spending with a denominator covering both

fee-for-service and Medicare Advantage where feasible.

8 Relation to the literature

This study connects two literatures. The first examines interior immigration enforcement. Secure Communities increased the link between local arrest and federal removal. Almuhausen, Amuedo-Dorantes, and Furtado (2024) find that its rollout increased institutionalization among older Americans by 0.26 percentage points, or 6.8 percent, and provide suggestive evidence of reduced work hours among housekeepers, personal-care aides, and home-health workers. Their evidence makes the care market a natural place to look for spillovers from a policy that moves enforcement in the opposite direction.

The second literature studies sanctuary policies themselves. Hausman (2020) estimates that county non-cooperation reduced deportations without increasing crime. Subsequent work has examined crime, reporting to police, labor markets, and financial inclusion. Health research is newer and often cross-sectional or based on policy-content indices. Ortiz et al. (2021) document that health-relevant provisions are common in sanctuary policies, while Nieri et al. (2023) and recent county studies report mixed health associations. Our contribution is to use exact county adoption dates, administrative health spending and utilization, and a modern staggered-adoption estimator.

The contribution is distinct from showing that immigration enforcement affects immigrants' own health-care use. Medicare beneficiaries are overwhelmingly older citizens and lawful residents. The outcomes therefore capture spillovers onto a public insurance program and local care system. That externality is relevant to the social cost of enforcement policy even if the policy's intended target population is not Medicare eligible.

8.1 Immigration enforcement and labor markets

Research on interior enforcement shows that policy affects much more than completed deportations. Enforcement programs change employment, geographic mobility, household structure, and the behavior of citizens living with noncitizens. These responses can spill into local product and labor markets. Care services are a particularly plausible margin because production is local, labor intensive, and difficult to automate.

The Secure Communities setting supplies the closest economic benchmark. Almuhausen, Amuedo-Dorantes, and Furtado exploit its staggered rollout and show increased institutionalization among older Americans, especially among groups likely to purchase household services. Their labor evidence points to reductions in hours among housekeepers, personal-care aides, and home-health workers. Our sanctuary-

policy design asks whether restricting local cooperation generates an opposite-direction adjustment and whether that adjustment reaches Medicare expenditures.

The two policy experiments are not symmetric. Secure Communities was a federal information-sharing program implemented across the country, while sanctuary rules are selected local responses to enforcement and litigation. The affected populations, salience, compliance, and counterfactual institutions differ. A positive effect of sanctuary adoption should not be interpreted mechanically as the negative of Secure Communities. The comparison instead motivates outcomes and establishes that elder-care markets are exposed to enforcement risk.

8.2 Sanctuary policies, crime, and institutions

Most quantitative sanctuary-policy work begins with public safety. The central empirical question is whether limiting detainer cooperation reduces removals at the cost of increased crime. Hausman finds a substantial reduction in deportations and no detectable increase in crime. Other county and city studies generally find null or crime-reducing associations, although designs and policy definitions vary. Evidence that victim reporting changes after sanctuary adoption suggests that institutional trust can be an outcome as well as a mechanism.

Economic studies extend the inquiry to employment, local economic activity, and financial inclusion. Protective policy can increase labor-market attachment and reduce the shadow cost of formal transactions for immigrant households. Those effects may raise local demand and tax bases as well as labor supply. For health markets, the relevant distinction is whether general economic improvement or care-specific labor adjustment drives the response. Our null establishment and wage findings do not resolve that distinction.

Policy measurement remains a common challenge. Sanctuary ordinances differ in legal coverage, enforcement, exceptions, and symbolic language. Ortiz and coauthors' content analysis shows that health-related language appears in more than two thirds of the policies they examine, including provisions concerning access, privacy, and trust. A binary detainer-policy indicator captures a coherent enforcement margin but not the broader policy environment. Heterogeneity by policy content is a natural extension once exact legal texts are coded consistently.

8.3 Protective immigration policy and health

The health literature on protective local policy remains small. Existing studies examine mental health, birth outcomes, safety-net participation, nutrition assistance, and medication use. Results are mixed, reflecting different populations, exposure definitions, and geographic scales. Nieri, Carney, and Kline report limited

evidence of improved Latinx mental health, while Goglichidze and coauthors find no statistically significant relationship between sanctuary environments and diabetes medication prescriptions among community-health-center patients.

Our outcome and population differ from most of that literature. Rather than focusing on immigrants' health, we examine county care markets and Medicare beneficiaries. This approach captures spillovers onto older residents who may be connected to immigrant workers or mixed-status households but are not themselves the direct target of enforcement. It also shifts the economic object from an individual health indicator to public insurance spending and utilization.

County-level analysis offers exact alignment with many sanctuary decisions and administrative outcomes, but it reduces the number of policy units relative to nationwide individual data. Hausman's sample intentionally focuses on 296 counties with large Hispanic populations, and half adopt. This design trades broad geographic coverage for a more relevant risk set and credible policy coding. The effective sample at long horizons is smaller still because only early cohorts can contribute.

8.4 Contribution to health economics

Health economics has long studied how household constraints, provider capacity, institutional access, and insurance design shape the location and intensity of care. Immigration policy belongs in this framework because it can change household behavior and local institutions without altering Medicare eligibility or reimbursement. The resulting incidence can reach patients, families, providers, and public insurers outside the population directly targeted by enforcement.

This paper's contribution is a reduced-form fiscal and utilization response. Local enforcement institutions predict Medicare spending several years later, and the accompanying rise in inpatient participation, days, and length of stay identifies where in the care pathway the response occurs. The null and inconsistent workforce tests are also informative: the spillover cannot be attributed confidently to a readily measured expansion of noncitizen direct-care labor. The remaining distinction among improved access, delayed severity, provider behavior, coding, and beneficiary selection is essential for welfare interpretation.

9 Limitations

Four limitations are material. First, adoption is concentrated in 2013 and 2014 and in California. The no-California model is smaller and only marginally precise. A state-by-year specification or border-county de-

sign would better absorb contemporaneous state changes, but the current treatment pattern may leave limited within-state identifying variation.

Second, beneficiary composition is not stable. Medicare Advantage participation, dual eligibility, fee-for-service enrollment, and population display differential pretreatment movement. Full enrollment files or beneficiary-level claims are needed to construct fixed-denominator outcomes and to determine whether the spending increase reflects utilization, prices not removed by standardization, or sample selection.

Third, no workforce source captures the complete relevant market. CBP and QCEW omit informal and private-household work and apply different coverage and suppression rules. The public ACS EEO tabulations add detailed occupation and citizenship but are pooled across five-year periods, impose county population thresholds, measure residence rather than workplace, and distinguish citizenship rather than documentation status. Their combined failure to corroborate a common capacity response limits the labor-supply interpretation but cannot rule out changes in hours, informal care, or household caregiving.

Fourth, policy adoption may bundle other local decisions and may respond to unobserved trends. Exact legal coding, state preemption and statewide sanctuary rules, local election timing, and enforcement measures should be incorporated. The Hausman deportation first stage is strong evidence about the policy's enforcement content, but the present merged panel does not re-estimate a county-year first stage over the health sample.

These issues determine the scope of the causal interpretation. The present evidence supports a reduced-form spending result in the study counties, while denominator stability, state-specific policy changes, and direct care-labor measurement remain priorities for further research.

10 Conclusion

County limits on immigration-detainer cooperation are followed by a delayed rise in standardized Medicare spending per beneficiary. A separate administrative analysis confirms that these policies materially disrupted the local jail-to-ICE pipeline: recorded declines rose and a much smaller share of detainees produced federal custody. The policy treatment therefore represents a consequential change in local enforcement practice rather than a nominal legal designation.

The Medicare response is delayed. The pooled estimate is approximately 2.2 percent after five years and 4.5 percent after eight years, has jointly insignificant pretreatment coefficients, and remains positive after omitting every treated state or treated county. The implied year-eight magnitude is about \$428 per baseline fee-for-service beneficiary in contributing treated counties. This translation is descriptive rather than

a national fiscal or welfare estimate. The utilization decomposition locates the change in a larger inpatient-user share, more covered inpatient days, and longer stays. These three outcomes have clean pretreatment profiles and survive within-family false-discovery-rate adjustment. Emergency visits, ambulance use, and acute readmissions do not increase detectably.

Neither of the principal candidate mechanisms supplies a complete explanation. QCEW workforce estimates are not reproduced in County Business Patterns, and public ACS data show no significant change in aides per older resident, noncitizen aide employment, or the noncitizen share of aides. Public outpatient measures are also mixed: broad outpatient participation and evaluation-and-management event counts fall modestly, testing increases, and the largest FQHC and imaging patterns begin before adoption. These findings make a simple workforce-expansion or broad-access story difficult to sustain.

The chapter therefore identifies two endpoints of a reduced-form sequence—a large change in local enforcement cooperation and a delayed change in Medicare inpatient utilization—without claiming that the first mechanically causes the second through an observed mediator. Additional inpatient care could reflect access, accumulated health needs, caregiver arrangements, provider behavior, coding, or beneficiary composition. Comparability-adjusted evidence is strongest for the 2014 cohort, while the 2013 cohort lacks adequate common support and remains sensitive to California. Beneficiary-level claims, a stable beneficiary cohort, or new externally imposed policy variation would be required to distinguish the behavioral pathways and support a broader welfare claim.

11 References

Callaway, B., and Li, T. (2019). Quantile treatment effects in difference in differences models with panel data. *Quantitative Economics* 10(4), 1579–1618. <https://doi.org/10.3982/QE935>.

Almuhaisen, A., Amuedo-Dorantes, C., and Furtado, D. (2024). Immigration enforcement and the institutionalization of elderly Americans. *Journal of Health Economics* 94, 102859. <https://doi.org/10.1016/j.jhealeco.2024.102859>.

Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, rdae007. <https://doi.org/10.1093/restud/rdae007>.

Callaway, B., and Sant’Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>.

de Chaisemartin, C., and d’Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110(9), 2964–2996. <https://doi.org/10.1257/aer.20181169>.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225(2), 254–277. <https://doi.org/10.1016/j.jeconom.2021.03.014>.

Hausman, D. K. (2020). Sanctuary policies reduce deportations without increasing crime. *Proceedings of the National Academy of Sciences* 117(44), 27262–27267. <https://doi.org/10.1073/pnas.2014673117>.

Nieri, T., Carney, M. A., and Kline, N. (2023). Sanctuary city policies and Latinx immigrant mental health. *SSM Population Health* 21, 101318. <https://doi.org/10.1016/j.ssmph.2022.101318>.

Ortiz, R., Farrell-Bryan, D., Gutierrez, G., Boen, C., Tam, V., Yun, K., Venkataramani, A. S., and Montoya-Williams, D. (2021). A content analysis of US sanctuary immigration policies: Implications for research in social determinants of health. *Health Affairs* 40(7), 1145–1153. <https://doi.org/10.1377/hlthaff.2021.00097>.

Goglichidze, S., Wang, W., Smith, L. H., Ezekiel-Herrera, D., Heintzman, J. D., Marino, M., Lucas, J. A., and Crookes, D. M. (2025). Sanctuary policies and type 2 diabetes medication prescription trends among community health center patients. *Health Affairs Scholar* 3(1), qxae178. <https://doi.org/10.1093/haschl/qxae178>.

Sun, L., and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics* 225(2), 175–199. <https://doi.org/10.1016/j.jeconom.2020.09.006>.

Chapter Appendices

S1. Formal care-sector outcomes

Home-health employment grows after adoption and reaches 10.6 log points in year five, equivalent to 11.1 percent. The profile is economically large and roughly monotone after year two. It does not, however, satisfy the baseline diagnostic: the six pretreatment coefficients are jointly significant ($p = 0.0016$), with positive early leads. A shorter four-lead model no longer rejects, but changing the policy-year assignment does not eliminate the long-window rejection. The timing therefore resembles continuation of a preexisting differential trend as well as a possible treatment response.

The remaining care-market outcomes do not supply corroborating evidence. Home-health establishments and annual pay have imprecise positive medium-horizon estimates and reject the six-lead joint test. Skilled-nursing employment is near zero or modestly negative after adoption. Skilled-nursing covered stays remain statistically indistinguishable from zero with a clean preperiod. The share of Medicare beneficiaries using home health rises slightly, but its pretreatment test rejects. Taken together, these outcomes cannot establish that sanctuary policy expanded formal home-health capacity or substituted care away from institutions.

The distinction between employment and establishments matters. Employment can grow through hiring at incumbent agencies, increased hours, or improved reporting, whereas an establishment response requires entry or the opening of additional locations. The point estimates are compatible with within-establishment adjustment, but their uncertainty is large. Annual pay also combines hourly wages, hours, and workforce composition, so a near-zero estimate does not rule out offsetting wage and hour changes.

The formal-sector coverage of County Business Patterns is another reason for caution. Many workers who help older adults are employed directly by households, work through arrangements classified outside home health, or supply unpaid family care. A sanctuary policy could affect this broader workforce without changing NAICS-coded home-health establishments. Conversely, agency employment could rise because demand changes for reasons unrelated to immigrant labor. The present data measure a market response, not the nativity or legal status of workers.

Figure S1. Sanctuary policies and home-health employment. Notes: Points are imputation difference-in-differences estimates. Bars show 95 percent confidence intervals with standard errors clustered by state. The six-lead joint pretreatment test rejects.

Figure S2. Sanctuary policies and skilled-nursing covered stays. Notes: Points are imputation difference-in-differences estimates. Bars show 95 percent confidence intervals with standard errors clustered by state.

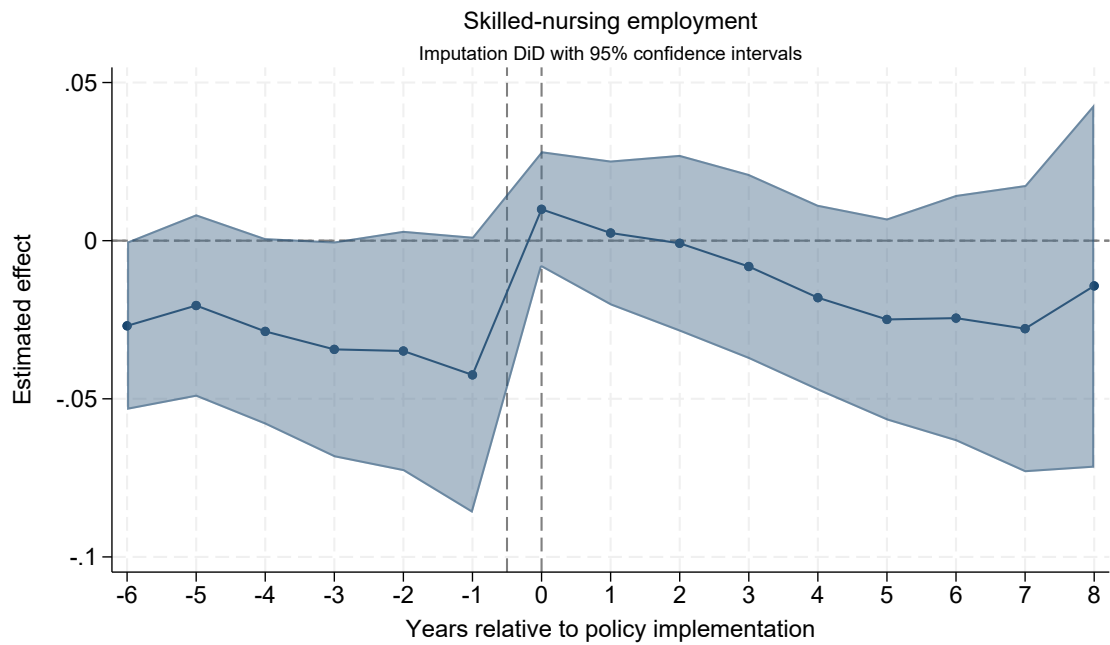
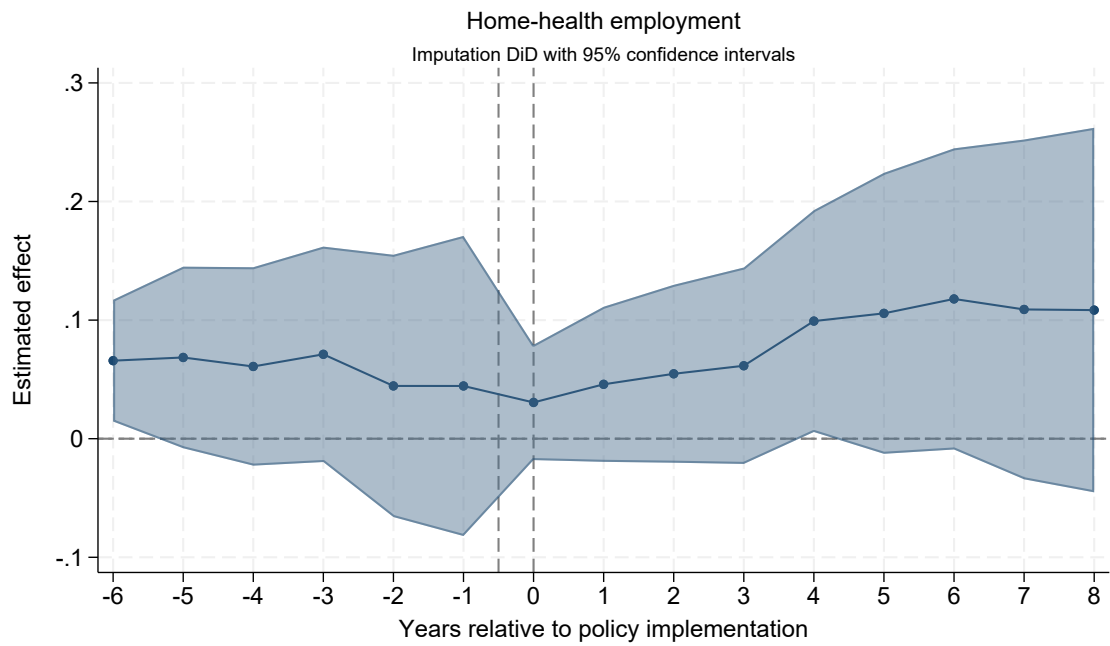


Table S1. Care-sector outcomes

Outcome	Year 5 percent	Year 8 percent	Pretreatment p	Observations
Home-health employment	11.15* (6.04)	11.45 (7.84)	0.002	4,647
Home-health establishments	3.45 (7.40)	6.05 (9.38)	0.030	4,647
Home-health annual pay	3.88 (2.70)	3.09 (2.61)	0.023	4,647
Nursing employment	-2.46 (1.63)	-1.42 (2.93)	0.025	4,526

Notes: Log coefficients are transformed to percentage changes; standard errors in parentheses use the log-point approximation multiplied by 100. Pretreatment p-values jointly test event times -6 through -1. Standard errors are clustered by state. *** p<0.01, ** p<0.05, * p<0.10.

S1A. Public ACS occupation and citizenship mechanism test

We independently test whether sanctuary adoption is followed by an expansion of the direct-care workforce using free public ACS Equal Employment Opportunity tabulations. The preperiod combines the 2006–2010 EEO-ALL02R, EEO-CIT02R, and EEO-NCIT02R county residence tables. The postperiod uses the 2014–2018 EEO-ALL02R and EEO-CIT02R tables, with noncitizen employment calculated as the difference between total and citizen counts. The primary harmonized occupation combines nursing, psychiatric, and home-health aides (code 3600 in the earlier release and 3601 in the later release). A broader sensitivity measure adds personal-care occupations, although its later category is less exactly harmonized.

The table’s population thresholds reduce coverage relative to the annual panel. We retain counties adopting in 2013 or 2014 and never-treated counties observed in both EEO periods. The balanced sample contains 242 counties, including 111 treated and 131 comparisons. Noncitizen outcomes contain 231 complete observations. The main change regression includes state fixed effects and 2010 log population, age composition, poverty, income, unemployment, hospital capacity, nursing-bed capacity, and the preperiod core-aide rate. Noncitizen models additionally include the preperiod noncitizen share. Standard errors are clustered by state.

Table S1A. Public ACS direct-care workforce estimates

Outcome	Estimate	Cluster SE	<i>p</i>	BH <i>q</i>
Log core aides	0.0878	0.0353	0.017	0.172
Core aides per 1,000 population	0.7697	0.4963	0.129	0.644
Core aides per 1,000 age 65+	3.1064	3.5696	0.389	0.738
Noncitizen share of core aides	0.0095	0.0136	0.490	0.738
Log noncitizen core aides	0.1674	0.1825	0.365	0.738
Log broad direct-care employment	0.0384	0.0586	0.516	0.738
Broad direct care per 1,000 population	0.2827	0.7443	0.706	0.797
Broad direct care per 1,000 age 65+	-0.1471	4.5154	0.974	0.974
Noncitizen share, broad direct care	0.0119	0.0114	0.300	0.738
Log noncitizen broad direct care	0.0582	0.1599	0.718	0.797

Notes: Estimates are treated-minus-comparison differences in the change between the 2006–2010 and 2014–2018 EEO tabulations. Models include state fixed effects and predetermined 2010 covariates. Standard errors are clustered by state. Benjamini–Hochberg *q* values adjust across the ten listed outcomes.

The log count of core aides rises by 0.0878, with a conventional clustered *p* value of 0.017. Its within-state permutation *p* value is 0.052, and its ten-outcome *q* value is 0.172. Overlap weighting produces an estimate of 0.0906; population weighting produces 0.0615 and is imprecise; excluding California produces 0.0856. The rate per older resident remains statistically indistinguishable from zero across these specifications. The cohort split is also uneven: log core aides change by -0.003 in the 2013 cohort and 0.089 in the 2014 cohort.

The direct immigrant-workforce predictions receive no support. The change in log noncitizen core aides

is imprecise, as is the change in noncitizen share. Overlap weighting reduces absolute standardized differences to essentially zero for the variables in the propensity model and to 0.015 for the held-out preperiod noncitizen share, without altering these conclusions. Because citizenship is not documentation status, the estimates cannot isolate undocumented workers. They do reject a simple account in which sanctuary adoption produces a large, readily detectable increase in noncitizen aides at the county level.

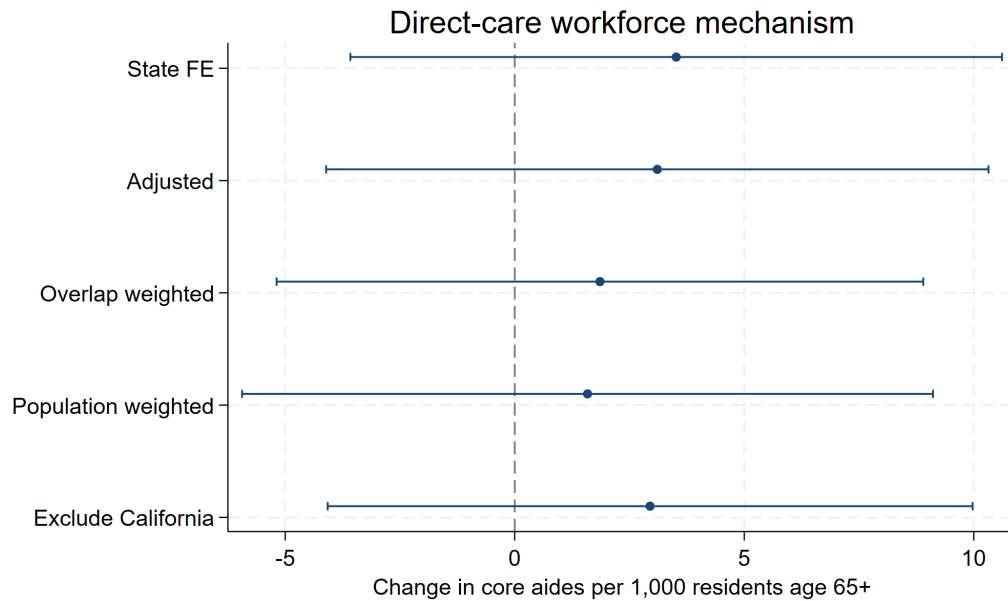


Figure S1A. Robustness of the change in core aides per 1,000 residents age 65 or older. Points are treatment coefficients; bars are 95 percent confidence intervals with state-clustered standard errors.

The EEO data are pooled five-year repeated cross sections rather than annual observations. The 2014–2018 window partially contains the adoption year for the 2014 cohort, residence may differ from workplace, and county thresholds change the estimand. We therefore use this exercise as an independent medium-run mechanism test rather than an event study. Together with disagreement between QCEW and CBP, its results prevent a workforce mediation claim.

S2. Beneficiary composition and denominator checks

Table S2. Composition and denominator checks

Outcome	Year 5	Year 8	Pretreatment p	Observations
Average age	-0.07 (0.11)	0.00 (0.14)	0.143	4,989
Dual eligible share	0.03*** (0.01)	0.04*** (0.01)	0.018	4,989
Medicare Advantage rate	-0.07*** (0.02)	-0.10*** (0.02)	0.013	4,989
Log county population	-1.98 (1.40)	-3.30** (1.69)	0.022	4,993

Notes: Shares are expressed as proportions in the source data and remain in proportion points. Log population is shown as a percentage change. Parentheses contain standard errors. All models use the baseline timing and six leads. *** p<0.01, ** p<0.05, * p<0.10.

S3. Distributional effects

The mean event study can conceal effects concentrated at particular points of the outcome distribution. We therefore estimate panel quantile treatment effects using the Callaway and Li (2019) change-in-changes design. The analysis is deliberately cohort specific. Counties first treated in 2013 are compared only with counties that are never treated during the panel, using 2011 and 2012 as the two untreated preperiods. Counties first treated in 2014 are compared with the same never-treated group, using 2012 and 2013 as the preperiods. For each cohort we estimate effects at horizons 0, 3, 5, and 8 and at deciles 0.10 through 0.90. Inference uses 499 county-level bootstrap replications.

This design asks whether the distribution of the treated cohort's change from the prepolicy path differs from the counterfactual distribution constructed from never-treated counties. It therefore complements the imputation event study rather than replacing it. The identifying restriction is distributional: absent treatment, treated counties' potential-outcome changes would follow the same conditional rank-based evolution as the comparison counties. The two-preperiod formulation allows the estimator to difference out the preceding change before comparing the postperiod outcome.

Table S3. Median Callaway–Li panel quantile treatment effects

Outcome	Cohort	$h = 0$	$h = 3$	$h = 5$	$h = 8$
Log standardized spending	2013	-0.056 [-0.092, -0.021]	-0.031 [-0.071, 0.010]	0.015 [-0.023, 0.052]	0.053 [0.004, 0.103]
Log standardized spending	2014	0.001 [-0.029, 0.030]	0.033 [-0.001, 0.067]	0.030 [-0.009, 0.068]	0.052 [0.016, 0.088]
Inpatient stays per 1,000	2013	-2.913 [-12.510, 6.685]	-3.091 [-18.765, 12.582]	8.073 [-7.534, 23.681]	15.099 [-4.099, 34.297]
Inpatient stays per 1,000	2014	-4.110 [-12.725, 4.505]	-5.439 [-17.493, 6.614]	-6.381 [-22.393, 9.631]	0.983 [-12.544, 14.510]

Notes: Entries are median QTT estimates with pointwise 95% confidence intervals in brackets. Log-spending effects are in log points; inpatient effects are covered stays per 1,000 beneficiaries. The 2013 cohort contains 40 counties and the 2014 cohort contains 86 counties through $h = 5$ and 82 through $h = 8$; the comparison group contains 148 never-treated counties. Each model uses $g - 2$ and $g - 1$ as the two preperiods and $g + h$ as the postperiod. Benjamini–Hochberg adjustments are calculated within each nine-quantile cohort–outcome–horizon family and across all 144 reported QTTs.

The spending distribution shows a delayed pattern near its center. For the 2013 cohort, the median effect is -0.056 log points at adoption, becomes small by year 5, and reaches 0.053 log points in year 8. For the 2014 cohort, the median effect is essentially zero at adoption and reaches 0.052 log points in year 8. Both pointwise year-8 median confidence intervals exclude zero, corresponding to about 5.5% and 5.4% increases. After controlling the false discovery rate across the nine quantiles in each family, the 2014 median remains significant at five percent ($q = 0.040$), while the 2013 median does not ($q = 0.080$). Across all 144 QTTs, the corresponding adjusted values are 0.091 and 0.265. The quantile results should therefore be read as stronger evidence for the 2014 cohort and suggestive replication for 2013, rather than as two independently definitive findings.

The inpatient distribution is less stable across cohorts. The 2013 cohort shows positive year-8 effects at

lower quantiles, including an estimated 35.7 additional stays per 1,000 at the 10th percentile, while its median confidence interval includes zero. The 2014 cohort instead shows positive year-8 effects around the 70th and 80th percentiles, again without a statistically precise median effect. This heterogeneity is consistent with a utilization response concentrated in different baseline-utilization segments across cohorts, but the absence of a common quantile pattern counsels against treating it as a single pooled mechanism.

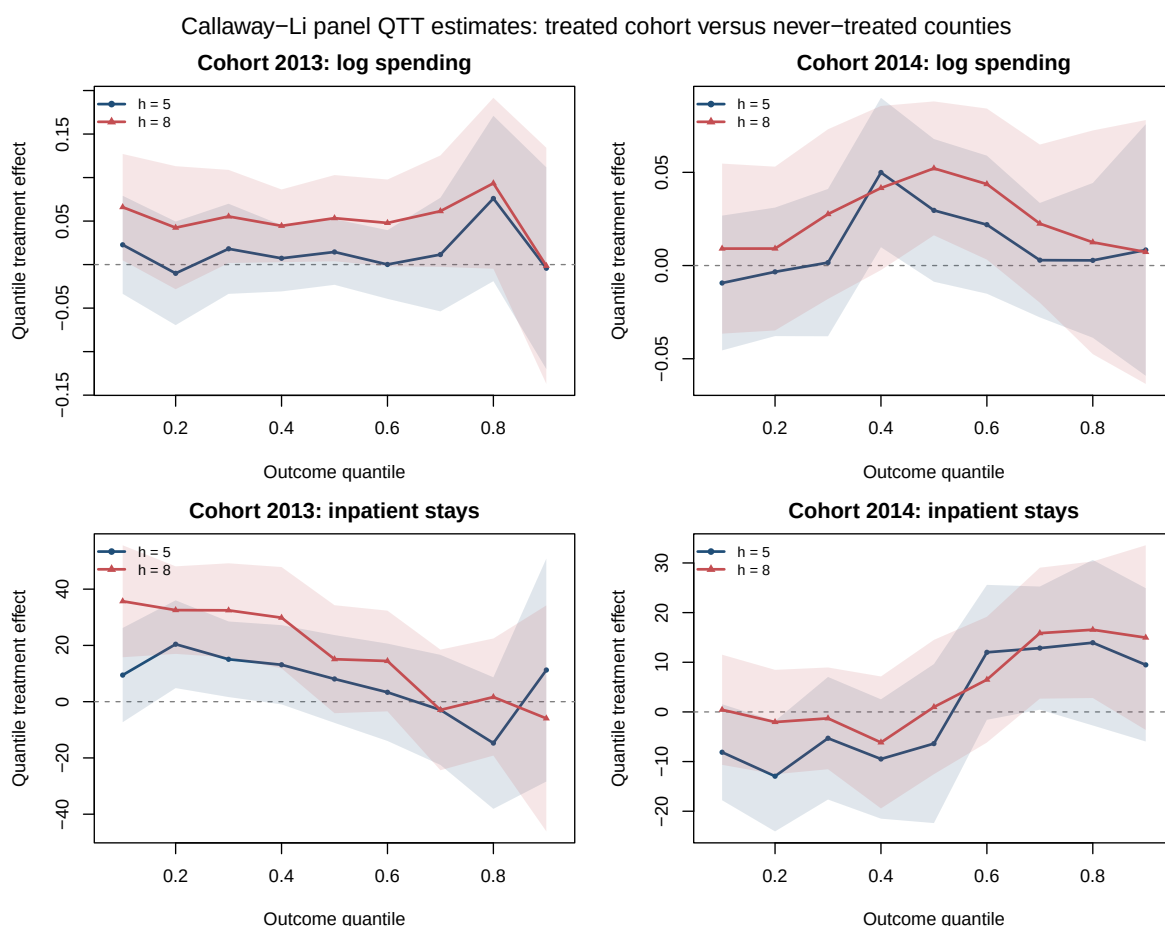


Figure S3. Callaway–Li panel quantile treatment effects at five- and eight-year horizons
Notes: Lines report decile-specific QTT estimates; shaded regions are pointwise 95% bootstrap confidence intervals. Each panel compares one treatment cohort with never-treated counties and uses the cohort-specific two-preperiod structure described in the text.

Taken together, the QTT analysis suggests that the delayed standardized-spending increase reaches the median in both large treatment cohorts, whereas inpatient utilization changes are distributionally heterogeneous and cohort dependent. The multiplicity-adjusted evidence is not equally strong across cohorts. Because the exercise uses only two large cohorts and never-treated controls, it is a targeted distributional diagnostic rather than an aggregate estimate for every treated county.

S4. Jurisdiction influence diagnostics

Table S4. Long-run spending estimates and influence diagnostics

Specification	$h = 5$	$h = 8$	Pretrend p	Counties	States
Full treated sample	0.022 (0.007)	0.044 (0.011)	0.183	296	43
2013 cohort versus never treated	0.027 (0.008)	0.075 (0.012)	0.000	188	33
2014 cohort versus never treated	0.027 (0.009)	0.037 (0.013)	0.005	234	40
Leave-one-state-out range	0.016–0.026	0.027–0.051	—	—	—
Leave-one-county-out range	0.021–0.024	0.043–0.047	—	—	—

Notes: State-clustered standard errors are in parentheses. Cohort rows retain the named cohort and never-ever-treated counties. All models use six placebo coefficients. Influence rows report the minimum and maximum point estimates across exclusions.

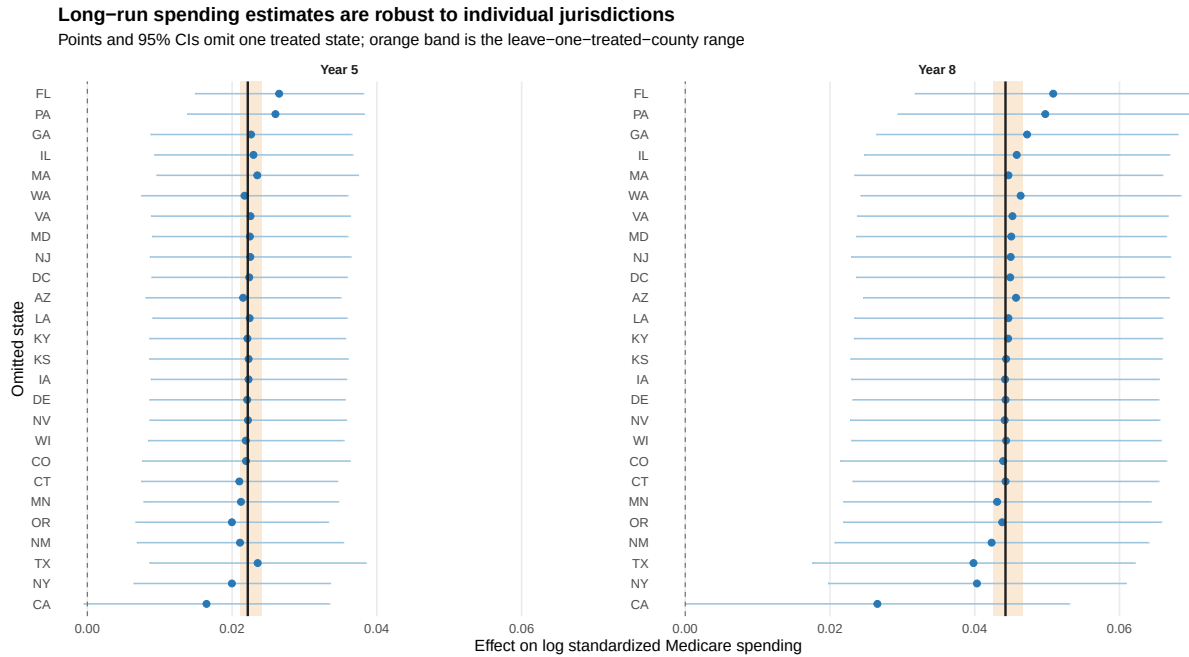


Figure S4. Leave-one-state and leave-one-treated-county influence audit

Notes: Points and horizontal lines report estimates and pointwise 95% confidence intervals after omitting the named treated state. The black vertical line is the full-sample estimate. The orange band is the range obtained by separately omitting each treated county.

S5. Additional access and utilization figure

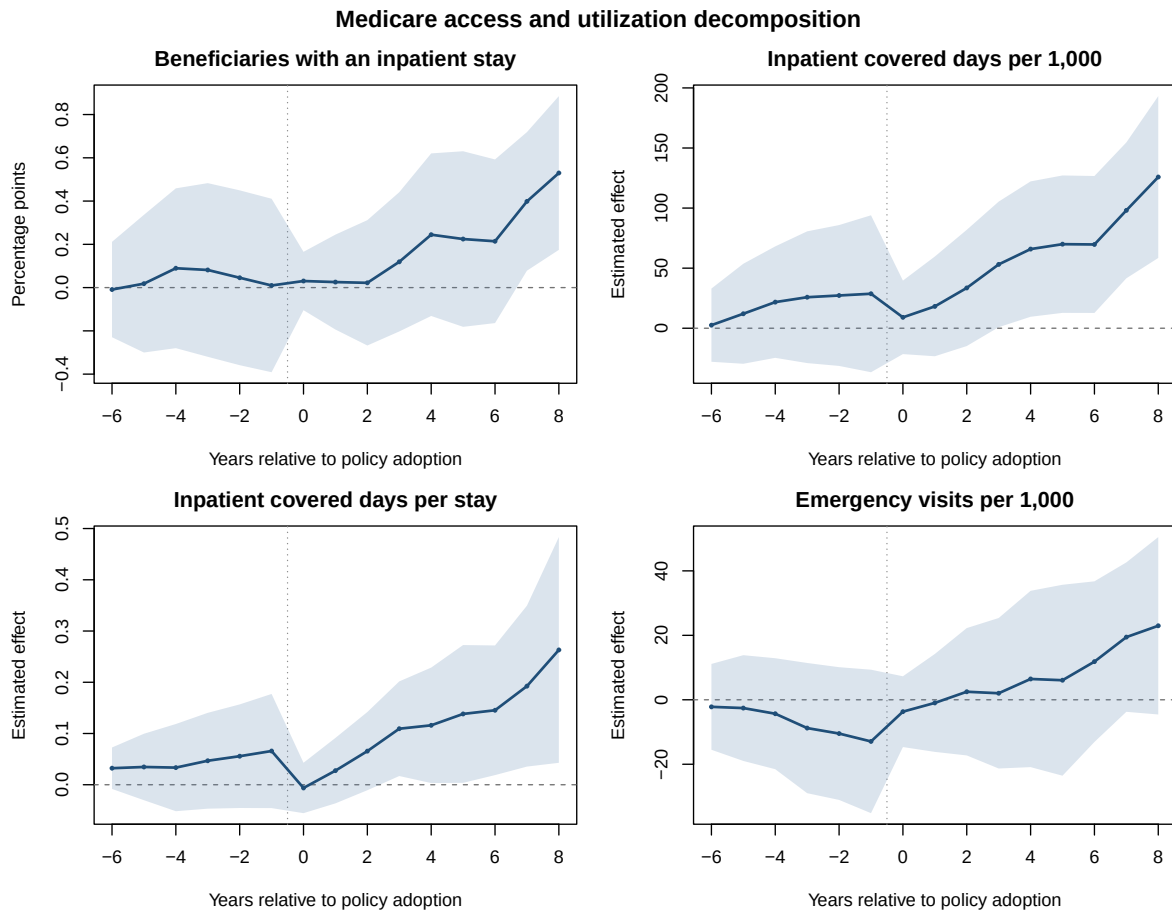


Figure S5. Medicare access and utilization decomposition

Notes: Points are imputation difference-in-differences estimates and shaded regions are pointwise 95% confidence intervals. Standard errors are clustered by state. The vertical line marks policy adoption.

S6. Additional online appendix

S6.1. Data sources and construction

Domain	Source	Construction and role
Policy dates	Hausman policies.dta	First qualifying county sanctuary-policy date; annual treatment year; five-digit county FIPS.
Care market	County Business Patterns	Employment, establishments, payroll, and annual pay for home-health and nursing-care industries.
Medicare	CMS county-level files	Standardized spending per beneficiary; inpatient and SNF covered stays; home-health use; beneficiary composition.
Occupation and citizenship	Public ACS EEO residence tables	Harmonized direct-care occupation counts for all, citizen, and noncitizen workers in 2006–2010 and 2014–2018.
Population and socioeconomic controls	Hausman archive, ACS, AHRF	Population, age structure, poverty, unemployment, foreign-born share, and health resources.

Notes: Raw source files, a download manifest with hashes, construction scripts, the processed panel, Stata do-files, logs, estimates, and coefficient-level CSV files are stored in the project directory.

S6.2. Event-time support

Event time	Medicare spending	Inpatient stays	Home-health employment
0	148	148	136
2	148	148	134
4	148	148	135
6	148	148	135
8	135	135	129

Notes: Entries are treated-county counts with raw outcome support at each event time. The estimator can use fewer observations when untreated fixed effects cannot be imputed. Long horizons increasingly reflect early adoption cohorts.

S6.3. Complete baseline coefficients for primary outcomes

Notes: Entries are coefficients with state-clustered standard errors in parentheses. Spending and employment are in log points. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S6.4. Reproducibility statement

The county-year analysis is implemented in Stata 18 using `did_imputation`. The processed panel is uniquely identified by county FIPS and year. The comprehensive do-file generates outcome-specific estimates, joint

Event time	Log Medicare spending	Inpatient stays	Log home-health employment
-6	-0.00 (0.01)	-0.73 (2.68)	0.07** (0.03)
-5	-0.00 (0.01)	0.27 (4.15)	0.07* (0.04)
-4	-0.00 (0.01)	1.73 (4.57)	0.06 (0.04)
-3	0.00 (0.01)	1.46 (4.93)	0.07 (0.05)
-2	-0.00 (0.01)	1.40 (4.93)	0.04 (0.06)
-1	0.00 (0.01)	1.27 (4.81)	0.04 (0.06)
0	-0.00 (0.00)	1.73 (1.63)	0.03 (0.02)
1	0.00 (0.00)	1.88 (2.53)	0.05 (0.03)
2	0.01 (0.01)	2.61 (3.27)	0.05 (0.04)
3	0.01** (0.01)	4.15 (3.47)	0.06 (0.04)
4	0.02*** (0.01)	6.17 (3.98)	0.10** (0.05)
5	0.02*** (0.01)	6.42 (4.26)	0.11* (0.06)
6	0.02** (0.01)	6.97* (3.94)	0.12* (0.06)
7	0.03*** (0.01)	11.28*** (3.44)	0.11 (0.07)
8	0.04*** (0.01)	14.02*** (3.90)	0.11 (0.08)

pretreatment tests, coefficient files, and event-study graphs; separate extension files generate timing, composition, weighting, California, and matched-risk-set checks. The public ACS EEO extension is implemented from reproducible download, construction, estimation, and randomization-inference scripts supplied with the submission package. All transformations begin with documented public or archived source files, and every reported estimate can be traced to a saved coefficient file or analysis log.

S6.5. Full outcome summary at medium and long horizons

Outcome	Year 3	Year 5	Year 8	Pretrend p	N
Home-health employment	6.35 (4.22)	11.15* (6.04)	11.45 (7.84)	0.002	4,647
Home-health establishments	-0.06 (5.90)	3.45 (7.40)	6.05 (9.38)	0.030	4,647
Home-health annual pay	1.97 (2.72)	3.88 (2.70)	3.09 (2.61)	0.023	4,647
Nursing employment	-0.81 (1.49)	-2.46 (1.63)	-1.42 (2.93)	0.025	4,526
Home-health beneficiary share	0.01* (0.00)	0.01* (0.01)	0.01* (0.01)	0.010	4,989
Skilled-nursing stays	-0.88 (1.67)	-0.89 (2.12)	0.95 (3.72)	0.682	4,989
Inpatient stays	4.15 (3.47)	6.42 (4.26)	14.02*** (3.90)	0.771	4,989
Standardized Medicare spending	1.35** (0.65)	2.24*** (0.68)	4.52*** (1.08)	0.183	4,989

Notes: Logged outcomes are transformed to percentage changes; level outcomes remain in source units. Standard errors appear in parentheses. The pretrend column gives the six-lead joint test. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S6.6. Treatment-timing sensitivity for primary outcomes

Notes: Spending and employment are transformed to percentages; inpatient stays remain per 1,000 beneficiaries. Standard errors appear in parentheses. Late-year timing rules assign policies in the named months to the next calendar year. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S6.7. Interpretation guide by outcome family

Timing rule	Outcome	Year 3	Year 5	Year 8	Pretrend p
Recorded policy year	Medicare spending	1.35** (0.65)	2.24*** (0.68)	4.52*** (1.08)	0.183
Recorded policy year	Inpatient stays	4.15 (3.47)	6.42 (4.26)	14.02*** (3.90)	0.771
Recorded policy year	Home-health employment	6.35 (4.22)	11.15* (6.04)	11.45 (7.84)	0.002
Shift July to December	Medicare spending	1.63** (0.68)	2.30*** (0.81)	4.91*** (1.49)	0.255
Shift July to December	Inpatient stays	4.79 (3.79)	6.89 (4.27)	15.08*** (5.07)	0.731
Shift July to December	Home-health employment	7.84 (4.88)	11.77* (6.71)	12.82 (8.01)	0.000
Shift October to December	Medicare spending	1.49** (0.70)	2.44*** (0.88)	4.97*** (1.65)	0.356
Shift October to December	Inpatient stays	4.53 (3.68)	7.27 (4.45)	15.64*** (4.91)	0.881
Shift October to December	Home-health employment	8.00 (4.82)	12.95* (6.72)	12.89 (7.76)	0.004

Outcome family	Role	Most useful evidence	Main limitation
Total standardized spending	Primary fiscal endpoint	Clean baseline pretrend; robust to annual timing	Composition and California sensitivity
Inpatient covered stays	Primary utilization endpoint	Clean baseline pretrend; positive long horizons	Long delay and changing cohort support
Inpatient participation, covered days, and days per stay	Primary mechanism family	Clean pretrends; long-run estimates survive within-family adjustment	Aggregate data do not distinguish access, severity, provider behavior, or coding
QCEW, CBP, and ACS workforce outcomes	Exploratory workforce mechanism	Three independent data systems, including occupation and citizenship tests	Results do not corroborate a common capacity or noncitizen-labor response
Establishments and annual pay	Provider entry and labor-price margins	Useful corroborating outcomes	Imprecise estimates and baseline pretrends
Skilled-nursing outcomes	Institutional and post-acute substitution	Clean utilization pretrend	Medicare SNF is not long-term residence
Home-health beneficiary share	Medicare home-health use	Directly connected to covered home health	Pretreatment rejection and limited mechanism support
Beneficiary composition	Denominator diagnostics	Reveals plan and eligibility trends	Cannot be controlled for mechanically after treatment

A Supplementary Appendix: Matched Risk-Set Design

A.1 Purpose and estimand

This appendix examines whether the delayed standardized-Medicare-spending result survives comparisons with observably similar counties. The exercise is cohort specific. For adoption cohort $g \in \{2013, 2014\}$, treated counties are compared exclusively with counties that remain untreated throughout the panel. The estimation sample covers event years -6 through 8 , and year -1 is the omitted reference period. The estimand is the average difference between the treated cohort and its weighted never-treated risk set at each event time.

The design addresses selection on observed pretreatment levels and trajectories. It cannot eliminate bias from unobserved shocks or policy determinants, and its credibility depends on overlap. We therefore report the complete balance and overlap diagnostics before interpreting the estimates.

A.2 Covariates and weighting algorithm

For each cohort, the balancing vector contains log standardized Medicare spending in every year from $g - 6$ through $g - 1$, the county-specific linear trend over those years, log population, nursing beds per 1,000 residents, Medicare Advantage participation, the average CMS risk score, and inpatient stays per 1,000 beneficiaries measured at $g - 1$. It also contains the age-65 share, poverty rate, log median household income, unemployment rate, and hospital beds per 1,000 measured in 2010. We use 2010 for the AHRF measures because those variables are not observed annually before both large treatment cohorts. No post-adoption covariate is used. A foreign-born share is unavailable in the panel before adoption and is therefore omitted rather than backcast from post-treatment data.

Let X_c denote the standardized covariate vector for never-treated county c and $\bar{X}_T$ the treated mean. Exact entropy balance would select positive weights that reproduce $\bar{X}_T$ while minimizing relative entropy. The exact constraints are infeasible for the 2013 cohort: the treated covariate vector lies outside the joint convex hull of never-treated controls. We therefore solve the regularized dual problem

$$\min_{\lambda} \log \left(\sum_{c \in C} e^{X_c' \lambda} \right) - \bar{X}_T' \lambda + \frac{\rho}{2} \lambda' \lambda, \quad w_c \propto e^{X_c' \lambda}.$$

The ridge parameter is selected from a fixed grid to minimize residual standardized imbalance subject to an effective control sample of at least 20. Control weights are normalized to sum to the number of treated counties; treated counties receive weight one. We also cap control weights at their cohort-specific 95th percentile and renormalize them as a sensitivity check. Propensity scores are reported only as an overlap

diagnostic and do not determine the primary weights.

Table SA1. Matched risk-set balance and weight diagnostics

Cohort	Treated	Controls	Effective controls	Maximum weight	Max. SMD , raw	Max. SMD , weighted	Share SMD ≤ 0.10
2013	40	148	21.5	6.99	1.113	0.281	0.59
2014	86	148	55.0	7.80	0.681	0.013	1.00

Notes: SMD denotes the standardized mean difference between treated counties and never-treated controls. Effective controls equal $(\sum w)^2 / \sum w^2$. The weighted SMD uses regularized entropy weights.

The weighting exercise exposes sharply different support across cohorts. For 2014, the maximum absolute SMD falls from 0.681 to 0.013; every included variable is below 0.10, and the effective comparison sample is 55.0 counties. For 2013, the largest SMD falls from 1.113 to 0.281, but population, income, unemployment, and beneficiary risk retain meaningful imbalance. The effective comparison sample is only 21.5 counties. The 2013 design therefore lacks the overlap required for a clean observables-adjusted comparison.

Covariate balance before and after regularized entropy weighting

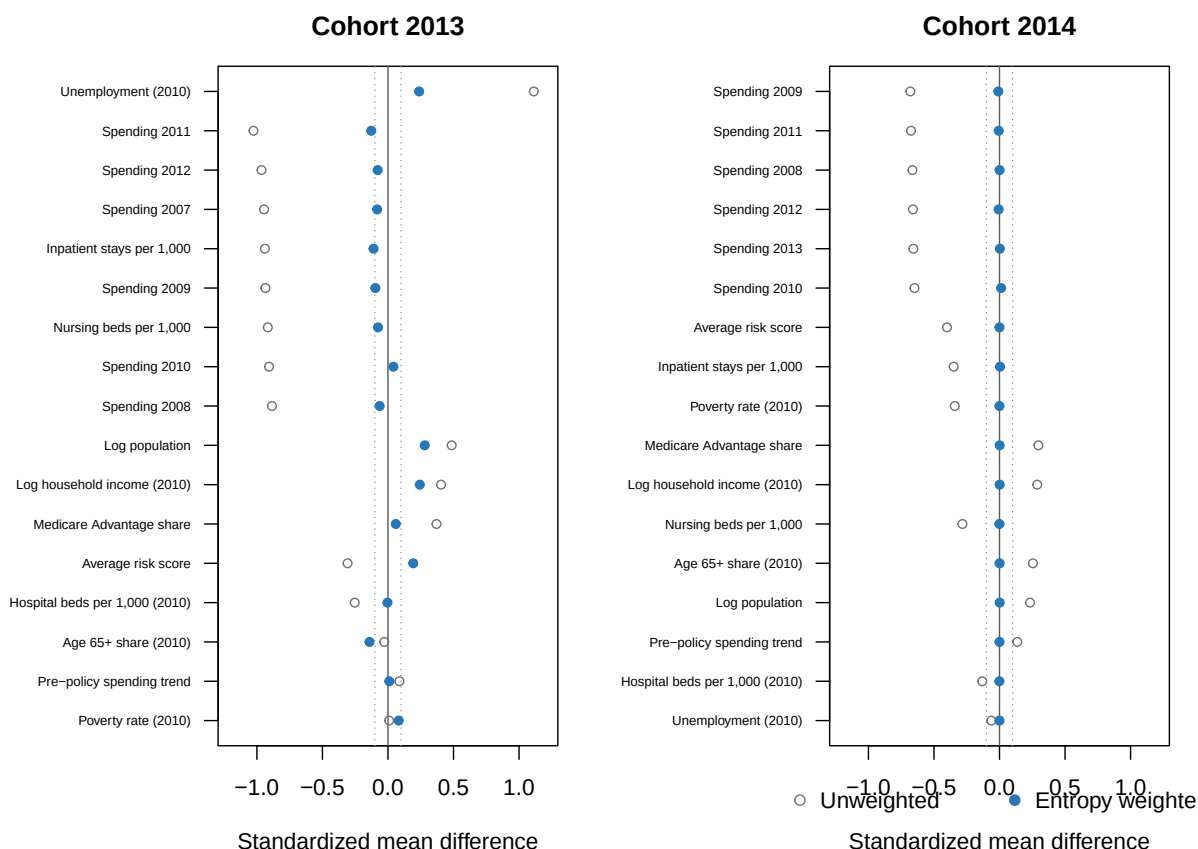


Figure SA1. Covariate balance before and after regularized entropy weighting

Notes: Open circles are unweighted standardized mean differences; filled circles are weighted differences. Dotted vertical lines mark ± 0.10 .

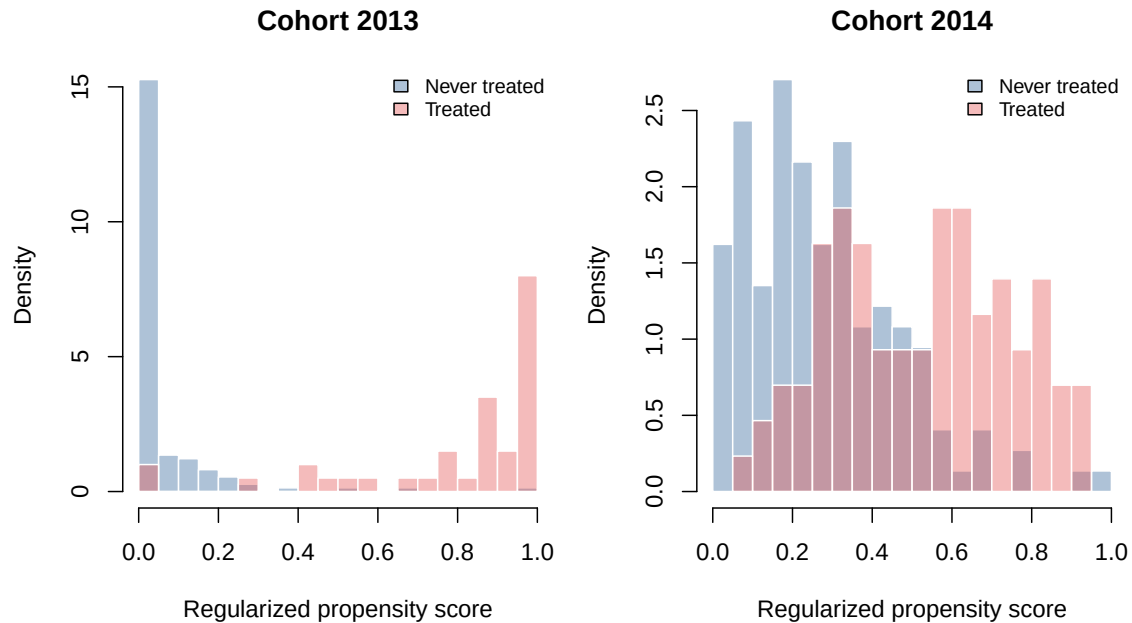


Figure SA2. Propensity-score overlap by adoption cohort

Notes: Histograms show regularized logit scores based on the full balancing vector. The separation for the 2013 cohort motivates treating its weighted estimates cautiously.

A.3 Weighted event-study specification and results

Within each risk set we estimate a weighted two-way fixed-effects event study with county and calendar-year fixed effects. Because all treated counties in a risk set adopt in the same year and all comparisons are never treated, this regression does not make already-treated counties serve as controls. Standard errors are clustered by state. The placebo test jointly tests event years -6 through -2 .

Table SA2. Matched risk-set estimates for log standardized Medicare spending

Cohort	Specification	$h = 5$	$h = 8$	Pretrend p
2013	Unweighted risk set	0.022 (0.010)	0.070 (0.014)	0.000
2013	Regularized entropy weights	0.007 (0.013)	0.047 (0.019)	0.000
2013	Weights capped at p95	0.024 (0.010)	0.063 (0.016)	0.000
2013	Entropy weights, excluding California	-0.063 (0.031)	-0.062 (0.042)	0.238
2014	Unweighted risk set	0.020 (0.009)	0.031 (0.014)	0.118
2014	Regularized entropy weights	0.019 (0.011)	0.022 (0.017)	0.829
2014	Weights capped at p95	0.024 (0.010)	0.030 (0.015)	0.526
2014	Entropy weights, excluding California	0.019 (0.011)	0.022 (0.017)	0.829

Notes: State-clustered standard errors are in parentheses. The omitted event year is -1 . The pretrend column reports the joint test for event years -6 through -2 . California exclusion retains the original cohort weights and removes California observations.

For the 2014 cohort, regularized entropy weighting eliminates the pretrend rejection ($p = 0.829$). The year-8 estimate remains positive at 0.022 log points but is imprecise ($p = 0.195$). Capping the weights at

the 95th percentile produces a 0.030 estimate ($p = 0.048$) with a clean joint pretrend test ($p = 0.526$). Thus, the 2014 evidence supports the sign and delayed timing of the main result, while precision depends on the weighting rule.

For the 2013 cohort, the weighted year-8 estimate is 0.047 log points ($p = 0.013$), but the placebo coefficients remain jointly significant ($p < 0.001$). Trimming produces a larger 0.063 estimate and also fails the placebo test. When California is excluded, the pretrend test no longer rejects ($p = 0.238$), but the year-8 estimate changes sign to -0.062 and is statistically insignificant. Given the limited overlap and remaining imbalance, the 2013 cohort cannot support a robust causal interpretation.

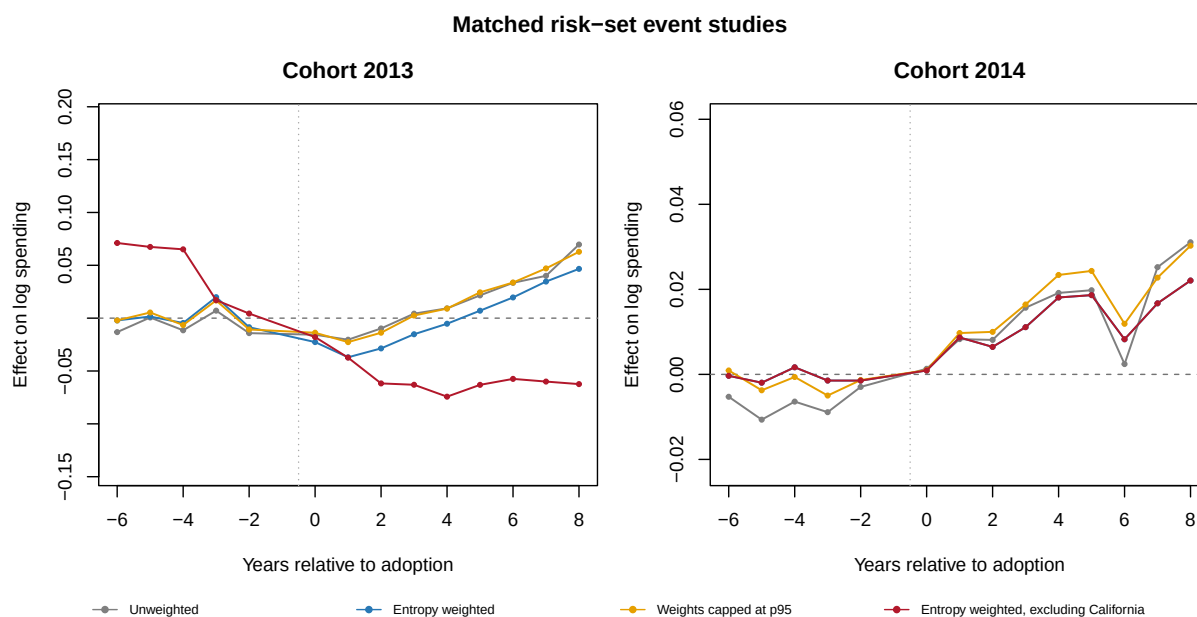


Figure SA3. Matched risk-set event studies

Notes: Lines report the unweighted risk set, regularized entropy weights, 95th-percentile-capped weights, and the entropy-weighted sample excluding California. Year -1 is normalized to zero. Confidence intervals are reported in Table SA2 and the replication files.

A.4 Interpretation

This exercise does not provide an unqualified validation of the pooled estimate. It produces a favorable balance and pretrend profile for the 2014 cohort, where the delayed estimate stays positive but loses precision under the primary regularized weights. It diagnoses poor common support for the 2013 cohort and substantial sensitivity to California. The combined evidence therefore supports a delayed positive association that is clearest for observably comparable 2014 adopters, while leaving a material identification concern for 2013 adopters. The pooled result should continue to be described as reduced-form evidence rather than a definitive

causal estimate.

All inputs and outputs are preserved. The script `prepare_matched_risk_sets.py` constructs weights and diagnostics from the main panel; `run_matched_risk_set_analysis.do` estimates every specification; and `plot_matched_risk_sets.R` reproduces Figures SA1–SA3. The generated county weights, matched panel, coefficient file, balance file, metadata, and Stata log are stored in the matched-risk-set replication directory distributed with this chapter.